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The Temporal Tuning of the *Drosophila* Motion Detectors Is Determined by the Dynamics of Their Input Elements

Highlights

- Input neurons to the fly motion detectors show diverse temporal filter properties
- Octopamine activation accelerates detector velocity tuning and input cell dynamics
- Dynamics of input neurons correctly predict velocity tuning of motion detectors

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In Brief

Recent algorithmic models of visual motion detection in the fly rely on processing of three input channels. By characterizing the spatiotemporal response properties of the neurons presynaptic to fly motion-sensing cells across two tuning states, Arenz, Drews, et al. predict functional roles for these neurons in computing the direction of motion.



The Temporal Tuning of the *Drosophila* Motion Detectors Is Determined by the Dynamics of Their Input Elements

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SUMMARY

Detecting the direction of motion contained in the visual scene is crucial for many behaviors. However, because single photoreceptors only signal local luminance changes, motion detection requires a comparison of signals from neighboring photoreceptors across time in downstream neuronal circuits. For signals to coincide on readout neurons that thus become motion and direction selective, different input lines need to be delayed with respect to each other. Classical models of motion detection rely on non-linear interactions between two inputs after different temporal filtering. However, recent studies have suggested the requirement for at least three, not only two, input signals. Here, we comprehensively characterize the spatiotemporal response properties of all columnar input elements to the elementary motion detectors in the fruit fly, T4 and T5 cells, via two-photon calcium imaging. Between these input neurons, we find large differences in temporal dynamics. Based on this, computer simulations show that only a small subset of possible arrangements of these input elements maps onto a recently proposed algorithmic three-input model in a way that generates a highly direction-selective motion detector, suggesting plausible network architectures. Moreover, modulating the motion detection system by octopamine-receptor activation, we find the temporal tuning of T4 and T5 cells to be shifted toward higher frequencies, and this shift can be fully explained by the concomitant speeding of the input elements.

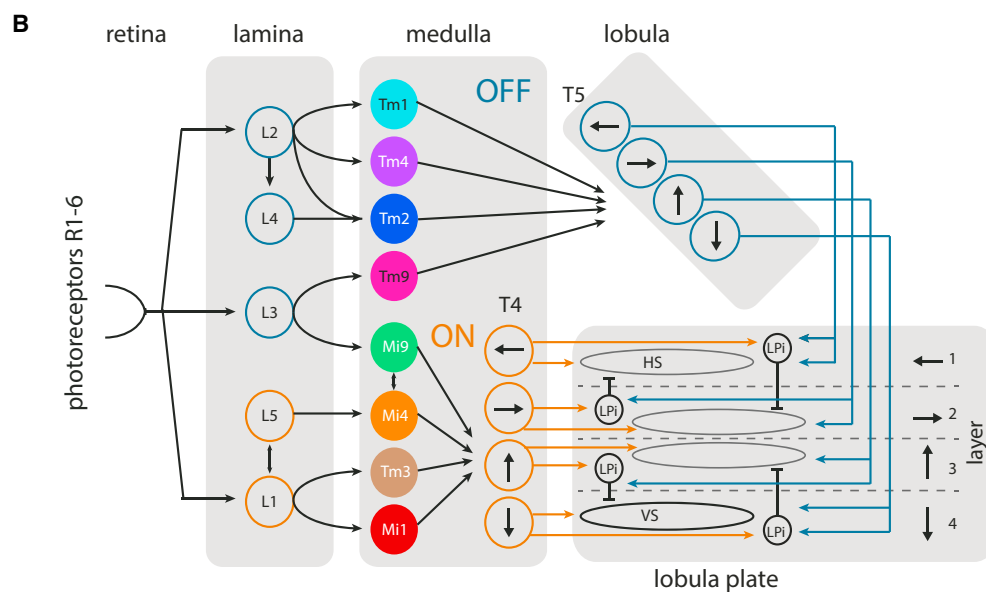
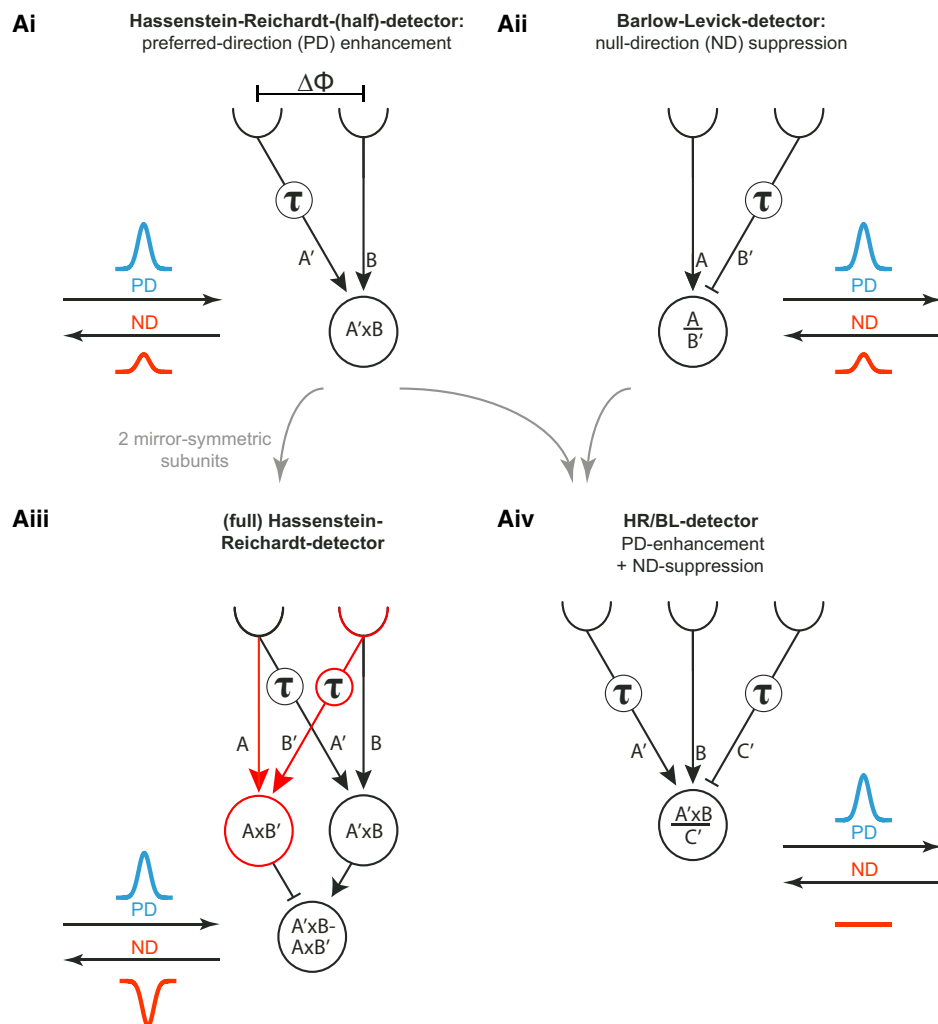
INTRODUCTION

The detection of visual motion arising from ego-motion is crucial for course stabilization in flies [1]. Sets of large tangential cells in the lobula plate of the fly optic lobe respond selectively to the optic flow resulting from whole-body rotation around different axes. As single photoreceptors respond to local luminance

changes in a non-direction-selective way, the intervening circuitry of the optic lobe [2–5] (Figure 1) must serve to extract the feature of visual motion by spatiotemporal comparison of the responses of neighboring photoreceptors.

Two competing algorithmic models of motion detectors have been proposed (Figure 1A). Both models rely on asymmetric temporal filtering of two input signals that are then fed into a non-linearity. They differ by the type of non-linearity employed and the location of the delay filter. In the Barlow-Levick (BL) detector (Figure 1Aii) [6], the delay is located on the preferred side and the non-linearity is inhibitory, leading to a suppression of signals moving in the null direction (ND). In the Hassenstein-Reichardt (HR) detector (Figure 1Ai) [7], the delay is located on the null side and the non-linearity is excitatory, leading to an enhancement of signals moving in the preferred direction (PD). In the full HR detector (Figure 1Aiii), two of those subunits, or half-detectors, are arranged in a mirror-symmetric fashion and subtracted from each other to yield a fully opponent detector (for review, see [8]).

How do the proposed elements of these algorithmic models map onto the neural circuits of the fly, and how does direction selectivity arise? The fly optic lobe consists of four neuropils downstream of the retina: the lamina, medulla, lobula, and lobula plate (Figure 1B). Photoreceptors synapse onto lamina monopolar cells. These lamina cells feed into two separate pathways encoding for different contrast polarities [9–11]: the ON pathway encodes brightness increments, and the OFF pathway encodes brightness decrements. In each pathway, the direction of visual motion is computed separately [12, 13]. In both pathways, lamina neurons connect onto a distinct set of medulla neurons. In the ON pathway, these medulla neurons have axon terminals in layer 10 of the medulla, where they overlap with the dendrites of T4 neurons [4]. In the OFF pathway, transmedulla neurons project to the lobula, where they synapse onto the dendrites of T5 neurons [5]. T4 and T5 neurons each fall into four subclasses, which respond selectively to visual motion in one of the four cardinal directions (front-to-back, back-to-front, up, and down) and project their axons according to this preference to one of the four layers of the lobula plate [14]. There, T4 and T5 cells converge and provide direct excitatory cholinergic input onto wide-field lobula plate tangential cells [15]. In addition, T4 and T5 cells synapse onto lobula plate intrinsic (LPI) neurons, which in turn inhibit tangential cells in the adjacent, oppositely tuned layer [16], making tangential cells fully motion opponent. Hence, T4 and T5



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neurons would represent the half-detector units of the fully opponent motion detector model just before the subtraction stage. Although the HR detector describes the responses of lobula plate tangential cells well, the responses of T4 and T5 neurons are more directionally selective than would be expected for the half-detectors of the HR model [14, 17].

In the ON pathway, medulla intrinsic neuron 1 (Mi1) and transmedullary neuron 3 (Tm3) were originally suggested as the main inputs onto T4 neurons from electron-microscopic reconstructions [4]. These data showed a small spatial offset of about a fifth of a column, about 1° in visual space, between Mi1 and Tm3 synapsing onto the same T4 neuron, with Tm3 located toward the null side of the T4 neuron. Based on this spatial offset, two possible implementations of the motion detector were suggested: a HR correlator with Tm3, or a BL detector with Mi1, as the delayed arm. Subsequent patch-clamp recordings showed a small temporal delay of ~ 20 ms for Mi1 with regard to Tm3, as well as a similar temporal offset of Tm1 with respect to Tm2 in the OFF pathway [18]. This led to the suggestion of HR correlator implementations with Mi1 and Tm1 as the delayed and Tm3 and Tm2 as the direct arms in the ON and the OFF pathway, respectively [18, 19].

However, new findings from several recent studies question this model. First, new electron-microscopic circuit reconstructions show additional synaptic input from Mi4 and Mi9 cells onto T4 cells (Lou Scheffer, personal communication; https://web.archive.org/web/20150218101857/http://emanalysis.janelia.org/flyem_tables.php), and from the transmedulla neurons Tm4 and Tm9 onto T5 cells [5]. Second, when all four input cell types in the OFF pathway were considered, large differences in their temporal response kinetics to flashes of dark bars were revealed [20]. Whereas Tm1, Tm2, and Tm4 respond like band-pass filters with different time constants, Tm9 has the response characteristic of a pure low-pass filter, together forming a filter bank that lends itself well to the construction of motion detectors. Third, whereas blocking the synaptic output of Mi1 severely reduces responses of tangential cells to moving ON edges, blocking Tm3 output only affects responses to edges moving at higher angular velocities but leaves responses to lower velocities unchanged [21]. This again argues against Tm3 being one of the two arms of the motion detector under all conditions. Similarly, in the OFF pathway, all four cell types were shown to contribute to the detection of moving OFF edges. Blocking their synaptic output decreased the responses of downstream tangential cells and

reduced the optomotor response to OFF edges [20]. However, no blocks of single cell types or of two types in combination fully abolished the responses to dark edges, suggesting either redundancy or a more complicated implementation than previously suggested. Fourth, recent experiments based on the sequential stimulation of individual laminar cartridges revealed that the elementary motion detectors in the ON pathway, T4 neurons, implement ND suppression [17] in addition to PD enhancement [22] (Figure 1Aiv). Spatiotemporal receptive fields of T5 neurons are consistent with a similar model in the OFF pathway [23]. This more elaborate motion detector implementation could explain the high direction selectivity. However, in contrast to both HR and BL detectors, it relies on at least three input elements.

Taken together, in both pathways, evidence mounts for a neural implementation that is more complicated than either the BL or the HR model alone, and there is a multitude of combinations possible to place the known columnar input elements into the proposed algorithmic three-arm model of the *Drosophila* motion detectors.

In order to dissect the roles and contributions of individual cell types, it would be helpful to modify their temporal response dynamics and observe the effect on the downstream motion detectors. One remarkable property of tangential cells is that their velocity tuning is not fixed but dependent on the behavioral state of the fly, as has been observed in *Drosophila* and *Lucilia*. In walking [24] as well as in tethered flying flies [25, 26], the temporal-frequency tuning shifts toward higher frequencies, corresponding to higher velocities, potentially matching the expected change of the stimulus statistics from resting to active locomotion. The behavioral effect can be mimicked in resting flies by pharmacological activation of octopamine receptors with octopamine [26] or the octopamine agonist chlordimeform (CDM) [25, 27]. The physiological source of this neuromodulation is octopaminergic neurons that project to the medulla, lobula, and lobula plate [28, 29]. They become activated during flight and are both necessary and sufficient for the increase in responses to higher temporal frequencies [26]. Importantly, this change in the temporal tuning could be reproduced in computer simulations by decreasing the low-pass filter time constant in the HR detector [25], indicating that identifying the input elements that change their kinetics under octopamine activation might help to pinpoint their functional roles in the detector.

Figure 1. Theoretical Models for Visual Motion Detection and the Underlying Neuronal Circuitry

(A) Algorithmic models of motion detectors based on variations of a common theme of spatiotemporal correlations of local luminance changes detected by photoreceptors. (Ai) In the Hassenstein-Reichardt (HR) correlator (of which a half-detector is shown here), a delay (τ) on the first of two arms activated by motion in the preferred direction (PD) causes coincidence of the two signals from neighboring photoreceptors (separated by an angle, $\Delta\phi$). A multiplicative non-linearity results in a PD enhancement. (Aii) In the Barlow-Levick (BL) detector the delay is located on the opposite arm, and the non-linearity is suppressive/inhibitory, causing a null-direction (ND) suppression. (Aiii) In the full HR correlator, two mirror symmetric subunits from (Ai) are subtracted, resulting in a fully opponent detector, which not only depolarizes in PD but also hyperpolarizes in ND. (Aiv) A recently proposed model, based on the responses of T4 neurons to apparent motion stimuli, combines PD enhancement and ND suppression along the PD axis.

(B) Schematic of the circuitry of the *Drosophila* optic lobe showing neuron classes suggested to be involved in visual motion detection. Local luminance changes are detected by photoreceptors in the retina and relayed via lamina monopolar neurons (classes L1–L5) and medulla neurons (Mi1, Tm3, Mi4, Mi9, Tm1, Tm2, Tm4, and Tm9) to T4 and T5 neurons. The latter are the first neurons in the visual pathway that respond selectively to motion. Both T4 and T5 form four subtypes that respond to one of the cardinal directions and project accordingly to the four layers of the lobula plate, thus forming a map of visual motion directions. In the lobula plate, they synapse onto large-field tangential cells (horizontal system [HS] and vertical system [VS] cells), as well as onto lobula plate intrinsic (LPI) cells that in turn form inhibitory synapses onto tangential cells in the adjacent layer of opposite PD. This inhibition corresponds to the subtraction stage in the full HR correlator (Aiii) and endows lobula plate tangential cells with full motion opponency.

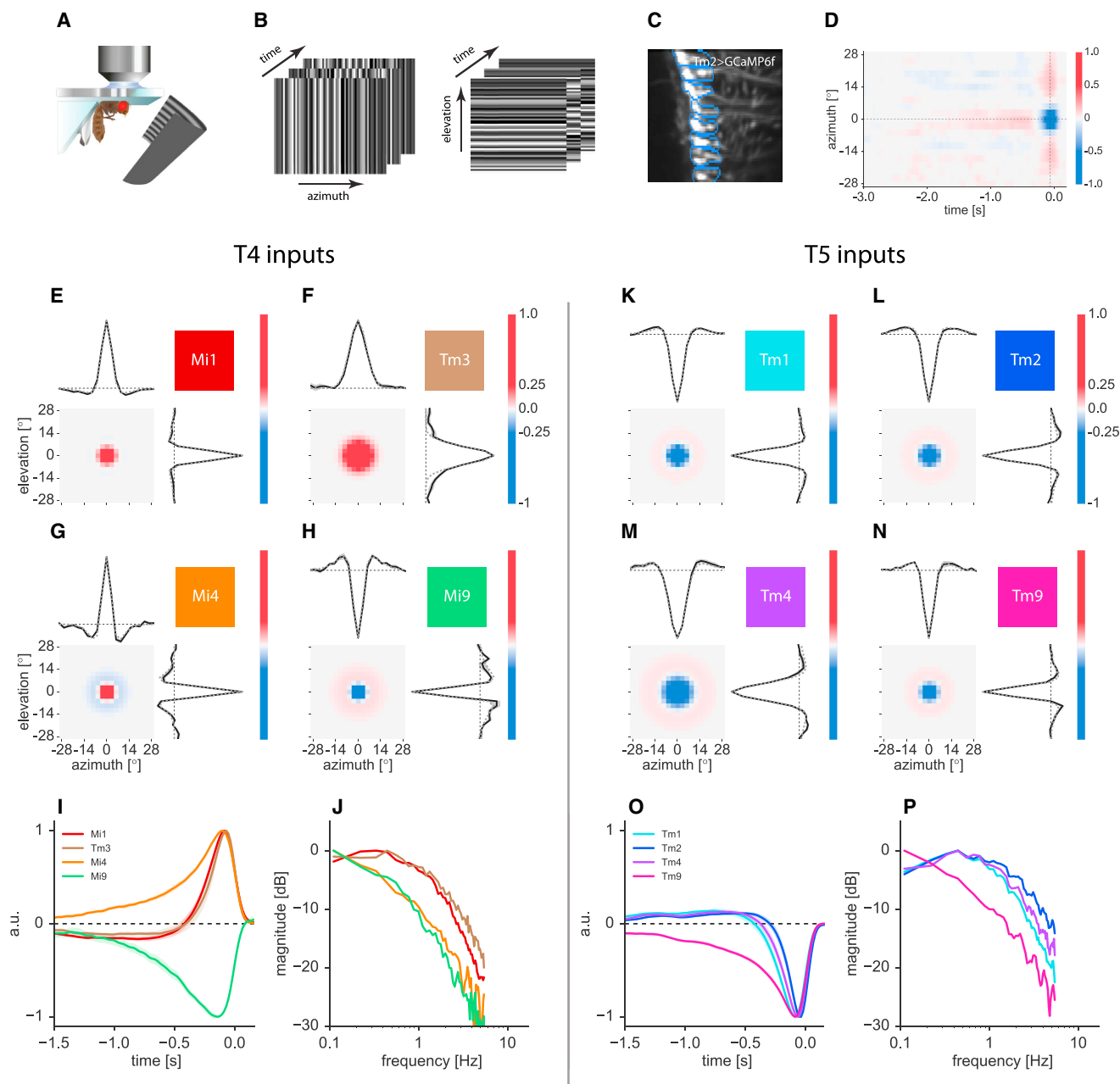


Figure 2. Response Properties of the ON- and OFF-Pathway Medulla Columnar Elements

(A) Two-photon calcium imaging of immobilized flies.

(B) Schematic of vertical (left) and horizontal (right) white-noise stimulus illustrated by three frames.

(C) Terminals of Tm2 neurons expressing the genetically encoded calcium indicator GCaMP6f. Regions of interest (ROIs) for the analysis of calcium indicator fluorescence changes encompass single terminals.

(D) Average aligned spatiotemporal receptive field of all Tm2 cells from (C) for a white-noise stimulus consisting of vertical bars. Along the vertical axis, the center-surround structure of the OFF-center receptive field is visible in the heat color code (vertical dashed line at the time of the peak of the response). The section along the time axis through the receptive field center reveals the temporal response kernel.

(E–H) Receptive fields of Mi1 (E), Tm3 (F), Mi4 (G), and Mi9 (H) for vertical (upper left) and horizontal (lower right) white-noise bar stimulation. From these, the two-dimensional receptive fields were constructed as a two-dimensional difference of Gaussians (Supplemental Experimental Procedures).

(I) Temporal kernels resulting from the reverse correlation of the calcium response with the white-noise stimulus for Mi1, Tm3, Mi4, and Mi9.

(J) Temporal kernels in frequency-space (constructed from the temporal kernels in (I)) revealing Mi1 and Tm3 as band-pass filters and Mi4 and Mi9 as low-pass filters. (For the measurements of the spatial receptive fields: Mi1: N = 5 flies, n = 35 cells; Tm3: N = 6, n = 37; Mi4: N = 5, n = 33; Mi9: N = 7, n = 29. For the determination of the temporal kernels twice as many measurements, from the horizontal and vertical one-dimensional noise stimulus, could be used.)

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In this study, we comprehensively characterize the spatiotemporal response profiles of all known columnar input elements of both the ON and OFF motion detectors in the fruit fly *Drosophila melanogaster* and take advantage of the motion detectors' state-dependent tuning characteristics. Using computer simulations, we test which combinations of input elements result in the observed properties of T4 and T5 neurons and thereby narrow down their possible cellular implementation. In particular, we address the question of whether the response dynamics of the input elements are sufficient to yield realistic motion detectors, or whether additional mechanisms on the synaptic or dendritic level are required to further modify the dynamics of the input signals.

RESULTS

Characterization of the Columnar Input Neurons to T4 Cells

The functional role of the input neurons to the elementary motion detectors and their correspondence to elements of any detector model depend crucially on their spatiotemporal response characteristics. For this reason, we characterized the spatial extent of the receptive fields as well as the response dynamics of all putative input elements to the T4 and T5 cells: Mi1, Tm3, Mi4, and Mi9 in the ON pathway, and Tm1, Tm2, Tm4, and Tm9 in the OFF pathway. Expressing the genetically encoded calcium indicator GCaMP6f [30] with cell-type-specific Gal4-driver lines, we imaged calcium signals in single terminals in layer 10 of the medulla or the proximal lobula for the ON- and OFF-pathway elements, respectively.

To precisely map the receptive fields of the input elements, we used a one-dimensional white-noise stimulus consisting of 2.8° wide horizontal or vertical bars covering the full extent of the arena (Figures 2A–2D; Figure S1; Supplemental Experimental Procedures). The spatiotemporal receptive fields were then determined from the neuron's calcium response by reverse correlation. The spatial components of these are the one-dimensional horizontal and vertical projections of the underlying two-dimensional spatial receptive field of the cell. In all cases, they strongly resembled a difference of Gaussians (DOG; also called a "Mexican hat"). Because they were similar for both the horizontal and vertical dimensions, we fitted a two-dimensional DOG to reconstruct two-dimensional spatial receptive fields (Figures 2E–2H and 2K–2N). The temporal component of the spatiotemporal receptive field reflects the temporal filtering properties of the neuron (impulse response). The extracted temporal filters were validated by predicting held-out test sequences of neuronal responses from the stimulus for two example neuron types (Mi1 and Tm9) (Figure S2; see Supplemental Experimental Procedures).

All four cell types in the ON pathway, Mi1, Tm3, Mi4, and Mi9, showed locally confined receptive fields that appeared isotropic in the horizontal and vertical dimensions (Figures 2E–2H). Mi1, Mi4, and Mi9 cells revealed a receptive field center with a half-

width diameter of approximately 6°–7°, corresponding to about one optical column. In contrast, the receptive field center of Tm3 was substantially larger, with a half-width diameter of about 12°. Mi4 and Mi9, and to a lesser degree Mi1, also revealed a significant antagonistic surround, giving them spatial band-pass characteristics. This antagonistic surround had a half-width diameter of approximately 20° for both Mi4 and Mi9 (Table S1). Because the area and thus the volume under the curve are proportional to the square of the radius, the amplitude ratio of surround to center should equal the inverse of the ratio of the squares of their half-widths for the center and the antagonistic surround to cancel perfectly. Notably, this relation is fulfilled for both low-pass elements, and the integrals of their surrounds perfectly match their respective centers, thus predicting no responses to wide-field flicker stimuli. At the same time, the spatial band-pass filter enhances responses to edges within the visual scene. In the case of Mi1, the integral of the surround reached about 50% of the one of the center. For Tm3, surround inhibition was completely absent, such that those cells have a pure low-pass characteristic in the spatial domain.

The temporal component of the spatiotemporal receptive field centers yielded the impulse responses, which reflect the temporal filtering properties of the respective cell type. Mi1 and Tm3 showed band-pass filter characteristics, as can be seen in their biphasic impulse responses (Figure 2I) and in their response spectra (Figure 2J). In contrast, Mi4 and Mi9 appeared as pure low-pass filters (Figures 2I and 2J). Surprisingly, and in contrast to the other elements of the ON pathway, Mi9 showed the inverse contrast preference, with an increased calcium response to darkening in its receptive field center (OFF response). However, apart from the polarity, the time course and filter characteristics of Mi9 were very similar to those of Mi4 (Figures 2I and 2J). Thus, the four ON-pathway elements can essentially be grouped into two classes: two fast-transient cells (Mi1 and Tm3) and two slow-sustained cells (Mi4 and Mi9). Within each class, the cells' impulse responses revealed only small differences.

Characterization of the Columnar Input Neurons to T5 Cells

We next performed analogous experiments on the OFF-pathway elements Tm1, Tm2, Tm4, and Tm9. Mirroring the situation in the ON pathway, all four neurons of the OFF pathway had locally confined isotropic receptive fields (Figures 2K–2N). In agreement with previous reports [18, 20], they were all excited by luminance decrements. Accordingly, they revealed an OFF receptive field center. The receptive fields of all four cells also had an antagonistic surround component, giving them a spatial band-pass characteristic. In contrast to Mi4 and Mi9, however, the surround inhibition, with respect to the center, was weaker, which should render them more responsive to wide-field flicker. As a parallel to the ON-pathway elements, three of the neurons, Tm1, Tm2, and Tm9, showed a receptive field center with a half-width diameter of about 7°, whereas one element, Tm4, had a larger receptive field center, with a half-width diameter of approximately 10°.

(K–P) Characterization of the inputs to T5 cells in the OFF pathway. Spatial receptive fields of Tm1 (K), Tm2 (L), Tm4 (M), and Tm9 (N). Temporal kernels in the time (O) and frequency domain (P) for the four input elements in the OFF pathway. (Tm1: N = 8 flies, n = 71 cells; Tm2: N = 9, n = 93; Tm4: N = 5, n = 35; Tm9: N = 5, n = 32.)

Graphs depict the mean. Shaded areas around the line, where displayed, represent $\pm$ SEM. See also Figures S1 and S2 and Tables S1 and S2.

The half-width of the antagonistic surround amounted to about 25° for Tm1, Tm2, and Tm9 and to 35° for Tm4 (Table S2). As for Mi1, and in contrast to Mi4 and Mi9, the antagonistic surround strength for all OFF input elements reached about 50% of the center, as calculated above on the basis of the amplitude and half-width ratios.

As for the ON-pathway elements, we assessed the temporal filter dynamics by measuring the impulse responses within the receptive field centers (Figures 2O and 2P). This revealed a clear band-pass characteristic for Tm1, Tm2, and Tm4 with rather short low-pass time constants of about 100–270 ms. In contrast, the impulse response of Tm9 reflected a pure low-pass filter with a much longer time constant of about 500 ms. Within the group of band-pass filters, Tm1, Tm2, and Tm4 responses have different time courses (Figure 2O) and response spectra (Figure 2P), corroborating a previous study [20]. Thus, as a striking difference from the ON-pathway elements, where two fast and two slow cells are found, the OFF pathway comprises three fast and only one slow cell.

Application of the Octopamine Agonist CDM Changes the Temporal Frequency Tuning of T4 and T5 Cells

It has previously been shown that activation of the octopamine system modulates the temporal-frequency tuning of lobula plate tangential cells [25, 26]. This effect could be implemented directly at the level of the tangential cells, or indirectly, by modifying the temporal tuning properties of its presynaptic input neurons, i.e., the T4/T5 cells. The latter case would give a handle to manipulate the elementary motion detectors and potentially allow narrowing down of their cellular implementation.

We first confirmed that the activation of the octopamine system with the octopamine agonist CDM [31] at a concentration of 20 μM [25] shifts the temporal tuning of tangential cells in the lobula plate of immobilized *Drosophila* to higher frequencies (Figure S3), corroborating earlier findings using octopamine [26].

Next we focused on T4 and T5 neurons. We performed two-photon Ca^{2+} imaging in *Drosophila* expressing the genetically encoded calcium indicator GCaMP6m in the subset of T4/T5 neurons that are upward motion selective and project their axons to layer 3 of the lobula plate (T4c/T5c) (Figure 3A). Visual stimulation was presented on a semi-cylindrical LED arena and consisted of full-contrast square-wave gratings with a spatial wavelength of 24° , moving at 12 different velocities ranging from $1.2^\circ/\text{s}$ to $480^\circ/\text{s}$, corresponding to temporal frequencies from 0.05 to 20 Hz, in PD and ND. Responses of T4 and T5 neurons were quantified as relative change of fluorescence (DF/F) amplitudes within small regions of interest in lobula plate layer 3 (example traces in Figure 3B). We found a temporal-frequency optimum of 1 Hz for motion in PD (Figure 3C, black traces). Application of CDM shifted the temporal-frequency optimum from 1 Hz in control to about 2.5 Hz (Figure 3C, magenta traces). Recording Ca^{2+} signals from the dendrites of either T4 or T5 cells, we found that T4 and T5 cells, considered separately, exhibited a similar temporal-frequency tuning, under control conditions as well as after application of CDM, and a similar shift in their tuning with CDM (Figures 3D and 3E).

In order to distinguish changes in the response to isolated motion stimuli from changes in the temporal integration of periodic signals, we also tested the velocity tuning of T4 and T5 neu-

rons to moving edges. For this, we presented bright and dark edges of full contrast moving at different speeds ranging from $3^\circ/\text{s}$ to $300^\circ/\text{s}$ in PD (Figures 3F and 3G). Corroborating previous results [14], T4 neurons responded selectively to bright edges, whereas T5 neurons were found to be selective for motion of dark edges. Measuring the calcium responses in the axon terminals in the lobula plate, we found that under control conditions the responses were highest to edges moving at the slowest velocity of $3^\circ/\text{s}$ for both ON and OFF edges, i.e., T4 and T5 neurons, respectively (Figures 3F and 3G, black traces). As was seen for the grating stimuli above, application of CDM shifted the optimal stimulus condition to higher velocities of $12^\circ/\text{s}$ (Figures 3F and 3G, magenta traces).

Therefore, the shift of the temporal tuning properties of lobula plate tangential cells during flight or mimicked by the application of octopamine-receptor agonists (Figure S3 [25, 26]) is already present at the level of the T4 and T5 cells, thus affecting the tuning of the elementary motion detectors.

Octopamine-Receptor Activation Speeds the Input Elements of T4 and T5 Cells

Different possible mechanisms could explain this shift of temporal tuning in T4/T5 cells. On the one hand, octopamine signaling could affect the synaptic inputs onto T4 and T5 neurons by changing the kinetics of neurotransmitter receptors or the dendritic integration of those signals in T4/T5 neurons. Different input elements with different response kinetics could differentially contribute to the postsynaptic signals in different states through changes in their response amplitude or via their synaptic weight. On the other hand, the kinetics of some or all input elements could speed up. We set out to test the latter hypothesis, i.e., that the response characteristics and tuning of the elementary motion detectors result directly from the temporal dynamics of the respective input elements.

For this, we characterized the spatiotemporal receptive fields of all input elements in both the ON and OFF pathways after activation of the octopamine system with CDM and compared them to control conditions. Application of CDM left the spatial receptive fields of all four input neurons in the ON pathway unaffected (Figure 4A). However, it accelerated the response kinetics of all four cell types to different degrees, with much stronger effects on the fast band-pass elements Mi1 and Tm3 than on the slow low-pass filters Mi4 and Mi9 (Figures 4B and 4C, magenta traces; Figures S4A, S5Ai, and S5Bi). As for control conditions, response kinetics of Mi1 and Tm3, as well as of Mi4 and Mi9, remained similar to each other after addition of CDM. In the OFF pathway, the results were very similar. The spatial receptive fields appeared unchanged by CDM for any of the columnar input neurons (Figure 4D). However, in the temporal domain, addition of CDM to the bath sped up the impulse responses significantly (Figures 4E and 4F, magenta traces; Figures S4B, S6Ai, and S6Bi), as was seen in the ON-pathway band-pass elements.

Computer Simulations Based on the Input Elements' Temporal Filters Suggest Candidate Motion Detectors

The input elements to the motion-detecting neurons T4 and T5 can be roughly grouped into two classes: temporal low-pass filters with large time constants, and band-pass filters with

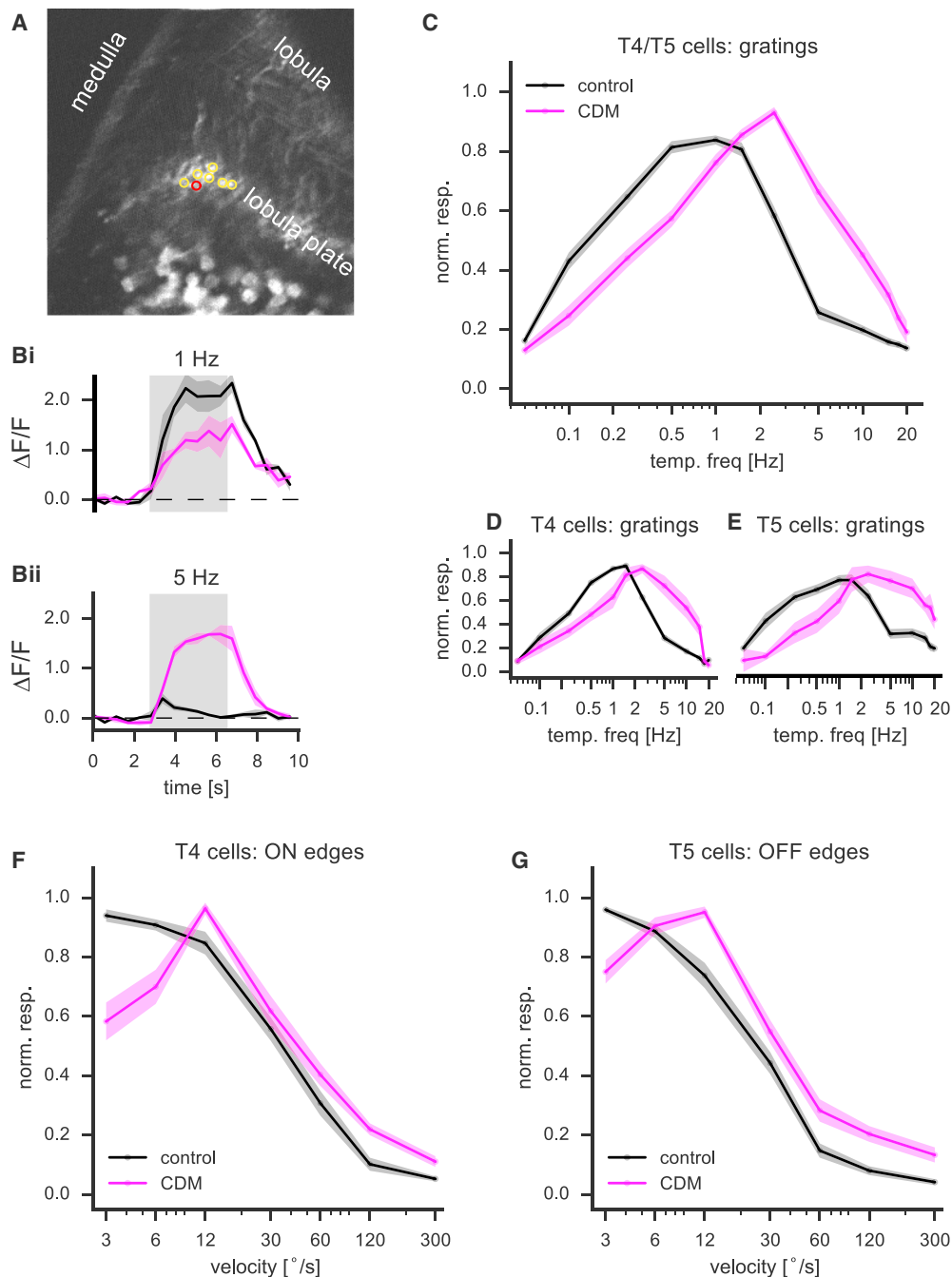


Figure 3. Application of CDM Shifts the Temporal-Frequency and Velocity Tunings of T4/T5 Cells to Higher Velocities

(A) T4/T5 neurons of the upward motion-selective subtype “c” projecting their axons to layer 3 of the lobula plate, expressing the genetically encoded calcium indicator GCaMP6m. The circles mark ROIs in the lobula plate; the red circle corresponds to the example calcium traces in (B).

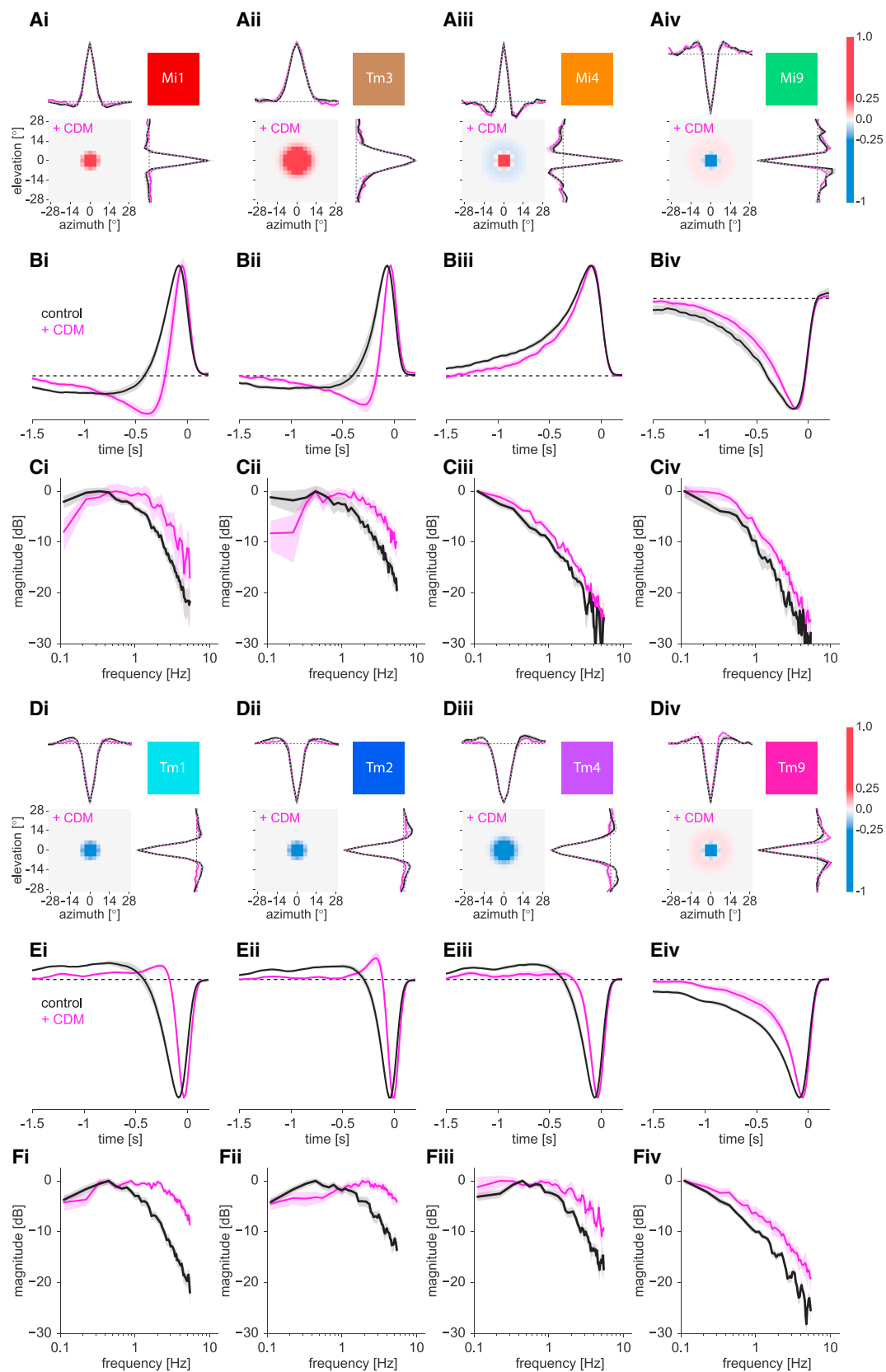
(B) Example of calcium responses (fluorescence changes) in the axon terminals of T4/T5 cells in response to square-wave gratings moving at temporal frequencies of 1 Hz (Bi) and 5 Hz (Bii) in control (black) and after application of CDM (magenta).

(C) Population average of responses of T4/T5 axon terminals to square-wave gratings moving in the PD (up). Application of CDM leads to a shift of the temporal tuning optimum ($N_{\text{cont}} = 36$ flies, $n_{\text{cont}} = 80$ ROIs; $N/N_{\text{CDM}} = 15/39$).

(D and E) Characterization of the temporal-frequency tuning in T4 (D) and T5 dendrites (E). As observed for the axon terminals, application of CDM (magenta) shifts the temporal-frequency tunings of both T4 and T5 cells to higher frequencies, as compared to control (black) (T4: $N_{\text{cont}} = 27$ flies, $n_{\text{cont}} = 52$ ROIs, $N/N_{\text{CDM}} = 9/14$; T5: $N/n_{\text{cont}} = 18/27$, $N/n_{\text{CDM}} = 7/9$).

(F and G) Population average of responses of T4 and T5 axon terminals in the lobula plate to bright (F; T4) and dark edges (G; T5), moving at different velocities, in control (black) and after application of CDM (magenta) ($N_{\text{cont}} = 9$ flies, $n_{\text{cont}_T4} = 21$, $n_{\text{cont}_T5} = 37$ ROIs; $N_{\text{CDM}} = 6$, $n_{\text{CDM}_T4} = 16$, $n_{\text{CDM}_T5} = 17$).

Graphs depict the mean. Shaded areas around the line represent $\pm$ SEM. See also Figure S3.



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significantly shorter time constants. We used the above-determined spatial receptive fields and response kinetics of the input elements and asked whether these could predict the responses of their postsynaptic targets, the elementary motion detector T4/T5 cells, without the necessity of additional filters or delays implemented either at the level of the synapses between the inputs and the T4/T5 cells or within the dendrites of the T4/T5 cells itself. In addition, we asked whether the observed shift in the temporal tuning in T4/T5 cells after application of the octopamine agonist CDM could be fully explained by the change of filter properties of the respective input neurons.

Although GCaMP6f has relatively fast kinetics when compared with other calcium indicators, it still possesses a decay time constant on the order of hundreds of milliseconds [30, 32]—long enough to significantly prolong the calcium signals of cells that have temporal dynamics on the same order of magnitude. In order to correct for this temporal filtering by the calcium indicator itself, we deconvolved the impulse responses in the frequency domain with a GCaMP6f low-pass filter (Figures S5 and S6). These corrected spectra were used as an approximation of the underlying filter properties of the input cells by fitting first-order filters to the average corrected frequency responses (Tables S1 and S2). We then used these values as well as the spatial filter characteristics in our computer simulations of a motion detector. Because the synaptic transmitters and postsynaptic receptors, and therefore the sign of the synaptic inputs, are not known, we decided not to make any assumptions about the sign of the synapses and ignored the response polarities of the determined receptive fields in our simulations.

Our simulations were based on a motion detector that combines PD enhancement and ND suppression, resembling a hybrid of a HR half-detector and a BL detector, as described in Haag et al. [17] (Figure 1Aiv). In this detector, three inputs with receptive fields offset by 5° each along the PD axis are processed such that an enhancing input A on the null side (left) forms a multiplicative non-linearity with the central, direct input (B), whereas a suppressing input (C) on the preferred side (right) implements a divisive non-linearity. The response of this detector equals the product of the input signals on the enhancing and the direct arm, divided by the signal from the suppressing arm (see the Supplemental Experimental Procedures).

There are 24 possible permutations that map the four input elements of each pathway onto the three positions of this detector, each one resulting in a detector with different tuning properties. Without making any further assumptions, we asked whether some of these combinations would yield more direction-selective

motion detectors than others. Each simulated detector was tested with moving square-wave gratings, and the responses were quantified in three ways (Figure 5A): (1) To assess how well the particular detector model discriminates between motion along PD and ND across velocities, we simulated square-wave gratings moving in PD and ND at different speeds covering more than three orders of magnitude. From the simulated responses, we calculated a direction selectivity index (DSI) as the relative difference between PD and ND responses, averaged over all grating velocities/temporal frequencies. (2) To judge the frequency tuning, we determined the temporal frequency evoking the maximum response in PD (temporal-frequency optimum, f_{opt}). (3) To characterize the direction tuning beyond PD and ND, emphasizing tuning sharpness, we simulated gratings moving in 12 equally spaced directions at the f_{opt} of each detector, as determined above. From those simulated responses, the normalized length of the tuning vector (L_{dir}) was calculated [33]. This tuning vector length of the hybrid detector was furthermore compared with the ones of the constituting HR and BL modules (Figures 1Ai and 1Aii, respectively).

In general, detectors with the low-pass filters Mi4 and Mi9 on both the outer enhancing and suppressing arms, flanking one of the band-pass elements Mi1 or Tm3, performed extremely well: they showed a rather high degree of direction selectivity and tuning sharpness (Figure 5B), in good agreement with the experimental data from T4 cells (compare with [14]), and their temporal-frequency optimum matched that of T4 cells as well (Figure 5B, right; compare with Figures 3C and 3D).

In addition, most combinations with one central low-pass neuron, Mi4, or, particularly, Mi9, flanked by the two band-pass elements Mi1 and Tm3, also achieved high direction-selectivity values. The PD (see arrows in Figure 5B, left) of these detectors is inverted as a consequence of the position of the delay in the HR and BL sub-modules. However, when considering both sub-modules separately (blue and red bars, respectively, in Figure 5B, right), the BL alone showed very low tuning sharpness (L_{dir}) and thus contributed little to the hybrid detector. This affects the tuning specificity of the hybrid detector, as can be seen when comparing, for example, Tm3xMi9/Mi1 with Mi9xTm3/Mi4. Both detectors are built on the same HR detector (using the same cells), but the one that employs Mi4 for the BL part of the model has a higher tuning sharpness. The same is true for all other pairs of this kind: given one pair of cells for the HR module, the implementation that places two low-pass filters on the outer arms of the detector always has the sharper tuning.

Figure 4. Activation of Octopamine Receptors Accelerates the Temporal Filters of the ON- and OFF-Pathway Medulla Columnar Elements (A) Spatial receptive fields of Mi1 (Ai), Tm3 (Aii), Mi4 (Aiii), and Mi9 (Aiv) for vertical (upper left) and horizontal (lower right) white-noise bar stimulation under control conditions (black traces) and after application of CDM (magenta traces and two-dimensional receptive fields).

(B) Temporal kernels for Mi1 (Bi), Tm3 (Bii), Mi4 (Biii), and Mi9 (Biv) revealing the faster time course after application of CDM (magenta) as compared to control (black).

(C) Temporal kernels in frequency-space, constructed from the temporal kernels in (B). Application of CDM (magenta) leads to a shift of the center frequency of the band-pass filters as compared to control (black). (For the measurements of the spatial receptive fields [controls are as in Figure 2]: Mi1: $N/n_{CDM} = 5$, $n_{CDM} = 31$; Tm3: $N/n_{CDM} = 6/34$; Mi4: $N/n_{CDM} = 5/38$; Mi9: $N/n_{CDM} = 7/37$. Again, the temporal kernel results determined from the horizontal and vertical one-dimensional noise stimuli were pooled, resulting in twice as many measurements.)

(D–F) Analogous to (A)–(C), the spatial receptive fields (Di–Div), temporal kernels (Ei–Eiv), and frequency spectra (Fi–Fiv) of the OFF-pathway elements Tm1, Tm2, Tm4, and Tm9. (Controls are as in Figure 2; Tm1: CDM: $N_{CDM} = 8$, $n_{CDM} = 67$; Tm2: $N/n_{CDM} = 9/93$; Tm4: $N/n_{CDM} = 5/28$; Tm9: $N/n_{CDM} = 5/42$.)

Graphs depict the mean. Shaded areas around the line represent $\pm$ SEM. See also Figure S4 and Tables S1 and S2.

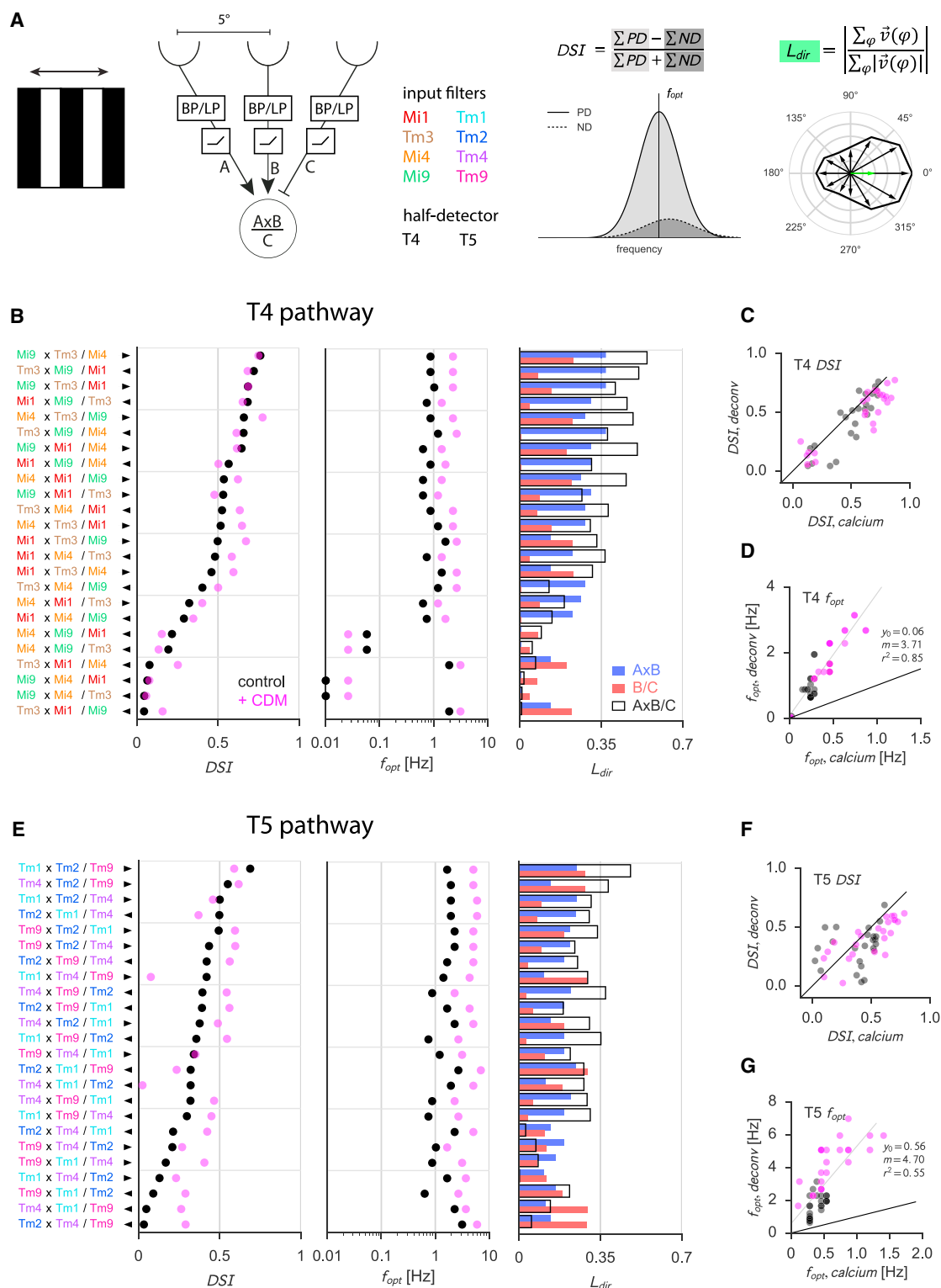


Figure 5. Computer Simulations of Elementary Motion Detectors

(A) Left: schematic of a three-arm detector combining a multiplicative PD enhancement and a divisive ND suppression. The positions of the enhancing (“A”), central (“B”), and suppressing (“C”) input can be occupied by any but different input elements. Those input elements are described by their temporal filtering characteristics, implemented as a band-pass (BP) and low-pass filter (LP) with subsequent rectification. The receptive fields of the three inputs are offset by 5° each. The simulated detectors are stimulated with square-wave gratings moving at different temporal frequencies in PD and ND. Middle: the direction selectivity (legend continued on next page)

Similarly, detectors that incorporated two elements with similar temporal response properties (such as Mi1 and Tm3) on two adjacent positions tended to perform worse, especially with respect to the L_{dir} value, indicating poor tuning sharpness. This can be easily explained by the fact that both the HR and BL modules of the hybrid detector rely on temporal differences in their respective two input arms. Inputs with more similar kinetics thus render the corresponding module less effective in creating direction selectivity. In fact, the best detectors were those where both halves showed high direction selectivities on their own (Figure 5B, right), provided the PDs of both modules were aligned.

Interestingly, almost all combinations showed a shift in their tuning toward higher temporal-frequency optima by about a factor of 2 when the filter properties after application of the octopamine agonist CDM were used, matching the experimental data. As a control that the direction selectivities in our simulations were not dependent on the used deconvolution filter, we repeated the simulations with the raw temporal kernels derived from the calcium responses. The same arrangements of input elements led to the motion detectors with the highest direction-selectivity values (Figure 5C), consistent with the notion that it is the relative filter properties that are crucial. Deconvolution merely changes the temporal frequency of the visual stimulus that leads to the maximum response (Figure 5D).

In the above simulations, we followed an unbiased approach with all inputs separated by 5° , thus having receptive fields arising from neighboring neuro-ommatidia. However, electron-microscopic reconstructions have shown a spatial offset between Tm3 and Mi1 cells projecting to the same T4 cell of about 1° in this order along the PD of the postsynaptic T4 cell [4]. The smaller spatial offset could counterbalance the small differences in temporal kinetics between these cells. Repeating the above simulations of the three-arm detector under these constraints still resulted in poorly direction-selective detectors for these combinations, with L_{dir} values of 0.38 (for Mi9xTm3/Mi1, as compared to 0.41 for a 5° offset) or less. In fact, when considering only a simple two-arm detector (HR or BL type), any detector that consisted of Tm3 and Mi1 with a spatial offset of 1° resulted in L_{dir} values of less than 0.06 for both types of detectors (in comparison to 0.13 for a 5° offset).

Although the evidence is weaker for the structure of the motion detector implementation in T5, we constructed analogous motion detectors for the OFF pathway with the measured receptive fields and response kinetics of the columnar inputs onto T5 neurons (Figure 5E). In contrast to the ON pathway, only one out of

the four input elements, Tm9, constitutes a low-pass filter, whereas the other three, Tm1, Tm2, and Tm4, exhibit band-pass characteristics. Most input element combinations resulted in motion detectors with low direction selectivity. Notably, the highest direction selectivity resulted from detectors with the low-pass filter Tm9 on the suppressing arm. Naturally, detectors with the fastest input (principally Tm2) in the central position flanked by two slower elements achieved higher direction selectivities, as with this arrangement the PDs of the HR and BL subunits are aligned. Arrangements with the sole low-pass filter, Tm9, in the central position resulted in detectors with poor directional tuning, both measured as DSI across all frequencies and L_{dir} , resulting from a virtually ineffective BL half (Figure 5E, right). Interestingly, combinations with the band-pass filters Tm1 and Tm4 constituting either half of the detector tended to perform comparatively poorly—and sometimes even showed a complete breakdown of direction selectivity—in at least one of the simulated physiological states. This can be explained by the fact that the small differences in the temporal response kinetics of these neurons were not stable between control and under CDM (Figure S6). As was seen for the ON pathway, using the spatiotemporal filters extracted under CDM in the simulations led to an increase of the temporal-frequency optimum by about a factor of 2 across all detectors (Figure 5E, middle, magenta dots). Again, the simulations were robust to the deconvolution applied to account for the filtering by the calcium indicator (Figure 5F). The best arrangements were the same irrespective of whether the raw or deconvolved filters were used, and only the temporal-frequency optimum was affected (Figure 5G).

Taken together, we find distinctly different response kinetics of the input elements in both the ON and the OFF pathway, from band-pass filters to pure low-pass filters. These map naturally onto hybrid elementary motion detectors implementing PD enhancement and ND suppression. The best-performing detectors arise when the fastest element occupies the central arm, flanked by slower inputs on the enhancing and suppressing arms. In the ON pathway, two low-pass inputs, Mi4 and Mi9, are found to fill this role. In the OFF pathway, the single low-pass element, Tm9, appears to be best positioned on the suppressing arm to achieve the highest direction selectivity.

DISCUSSION

To understand how motion detection is implemented on the dendrites of T4 and T5 cells, we describe in this study the response

of the detector is assessed across all temporal frequencies based on the area under the temporal-frequency tuning curves in PD and ND as the direction selectivity index: $DSI = (\Sigma PD - \Sigma ND) / (\Sigma PD + \Sigma ND)$. The dotted line indicates the temporal-frequency optimum (f_{opt}) for responses in PD. Right: illustration of the normalized tuning vector length (L_{dir}) as a measure for direction selectivity and tuning sharpness. L_{dir} is calculated as the vector sum of all responses according to the direction of stimulus motion, normalized to the sum of all response vector lengths.

(B–D) Characterization of the simulated motion detectors for the ON pathway.

(B) Direction selectivity (left), temporal-frequency optimum (middle), and normalized tuning vector length (right) for all possible permutations of the four ON-pathway input elements on the three positions of the simulated detector. The magenta dots indicate the effect of CDM application on direction selectivity and temporal-frequency tuning resulting from the accelerated temporal filters of the input elements. Arrows indicate the PD with respect to the corresponding cell arrangements. For the tuning vector length, the hybrid detectors (black open bars) were compared to their constituting HR (“AxB”; blue) and BL modules (“B/C”; red).

(C and D) Direction-selectivity indices (C) and temporal-frequency optima (D) of all detectors based on the deconvolved filter kernels as shown in (B) plotted against the detectors based on the raw calcium kernels.

(E–G) Same as (B)–(D) but for the OFF pathway.

See also Figures S5 and S6.

properties of the elementary motion detectors in *Drosophila*, the T4 and T5 neurons, as well as all of their known columnar synaptic input neurons, under two different tuning regimes. With this comprehensive characterization, we are able to narrow down the cellular implementation of the motion detectors and suggest probable wiring diagrams.

All of these input elements possess spatially restricted receptive fields with centers spanning one to two ommatidia. All, with the exception of the ON-pathway band-pass neuron Tm3, have pronounced antagonistic surrounds. Particularly for the low-pass filter elements Mi4 and Mi9, the strong antagonistic surround fully counterbalances the excitatory center. This should not only eliminate sensitivity to large-field flicker stimuli but more importantly curtail the otherwise tonic responses of pure low-pass filters to moving edges, and thus strongly improve direction selectivity. The locally confined receptive fields are in agreement with previous studies [18, 20, 34] but in contradiction to [35], which described Tm9 as a wide-field neuron. In both pathways, one neuron shows a larger receptive field (Tm3 in the ON pathway, and Tm4 in the OFF pathway). The larger receptive field sizes of Tm3 [18] and Tm4 neurons are consistent with the multi-columnar input these neurons receive based on electron-microscopic reconstructions [4, 5].

All elements of the OFF pathway respond to light OFF in the center of their receptive fields, consistent with [20]. In the ON pathway, Mi1, Tm3 [18], and Mi4 analogously show an ON-center response. Mi9, however, despite being an element in the ON pathway, responds positively to OFF stimuli. This could suggest a sign reversal through an inhibitory synapse onto T4. However, it is not known what neurotransmitter is released by Mi9, and thus whether it excites or inhibits T4 neurons.

Within each of the two pathways, we find a diversity of temporal filter characteristics from fast band-pass filters to pure low-pass filters with slow-sustained responses. These differences in temporal dynamics make them ideal components for motion detection without the need of postulating further processing by slow synaptic signaling or electrotonic filtering within the dendrites of T4 and T5 cells. Where the response kinetics of these cells have been previously described, our data are consistent. In particular, Mi1, Tm3, Tm1, Tm2, and Tm4 have previously been shown to respond transiently to sustained stimuli, i.e., to possess band-pass characteristics [18, 20, 34]. Tm3 appears faster than Mi1 [18] (but see [36]), and Tm2 faster than Tm1 [18, 20, 36]. However, these temporal differences are often very small. On the other hand, Tm9 in the OFF pathway has been described as a low-pass filter [20, 35], which matches our results. In the ON pathway, we find that the previously uncharacterized cell types Mi4 and Mi9 also show pure temporal low-pass response characteristics. Thus, in both pathways, input elements with slow-sustained and fast-transient responses are found, which then converge onto the dendrites of T4 and T5 cells, respectively. Yet the relative distribution differs. In the ON pathway, two input elements show pure low-pass characteristics (Mi4 and Mi9), whereas in the OFF pathway, Tm9 constitutes the only pure low-pass filter. Two of the three input elements that constitute pure low-pass filters, namely Mi9 in the ON and Tm9 in the OFF pathway, receive their lamina input primarily from the lamina monopolar neuron L3 [37]. As L3 has been shown to respond in a slower and more sustained fashion [38] than,

e.g., the transient L2 [10, 11], this could explain the low-pass characteristics of Mi9 and Tm9. L3, like all lamina neurons, responds positively to light decrements, and it releases the excitatory neurotransmitter acetylcholine, explaining the OFF response of Tm9 and Mi9. The response dynamics of Mi4 are likely to be heavily shaped by the strong reciprocal connections with Mi9 [37]. These reciprocal connections, and thus likely the cells themselves, would have to be inhibitory, as these cells show opposite response polarities.

Based on the spatial receptive fields and response kinetics, we could ask which input neurons could play which role in the motion detector. Previous computer simulations based on the measured dynamics of Tm cells in the OFF pathway have shown that most combinations of two elements result in classical (full) HR detectors with similar temporal tuning optima roughly matching the tuning of tangential cells [20]. In that study, only the combination of Tm2 and Tm4 could be excluded, as their filter time constants were too similar to each other to result in a functioning detector. However, subtraction of oppositely tuned half-detectors not only leads to motion opponency but increases direction selectivity of otherwise poorly tuned half-detectors. Consequently, the tuning of lobula plate tangential cells represents a rather indirect readout. By comparing simulations of the half-detector stage with recordings from T4 and T5 neurons, we can exclude the majority of possible combinations of input elements based on their temporal-frequency optimum or directional selectivity (see below).

Based on visual stimulation of single individual columns, T4 neurons have recently been shown to implement both PD enhancement and ND suppression [17]. The receptive fields of these interactions are spatially offset along the PD axis in this order. The corresponding hybrid of an HR half-detector and a BL detector requires a minimum of three columnar inputs: a fast central input, flanked by two outer inputs providing signals that are delayed relative to the central one.

In our computer simulations for the ON pathway (Figure 5), the majority of detectors with the highest direction selectivity fall into two groups: (1) the two low-pass filter elements Mi4 and Mi9 on the outer enhancing and suppressing arms, and either of the fast band-pass elements Mi1 and Tm3 on the central arm, matching the above layout, and (2) the inverted arrangement, with one central low-pass filter, flanked by the band-pass filter elements Mi1 and Tm3. This also resulted in an inverted PD.

In the latter case, however, the BL subunit considered alone contributed very little to the directional tuning (Figure 5B, right), as the low-pass-filtered central excitatory input tends to outlast the corresponding suppression from the band-pass outer arm. This reduces the tuning sharpness of these detectors. Furthermore, this implementation does not match the arrangement of PD-enhancement and ND-suppression receptive fields along the PD in this order found for T4 cells [17]. Additionally, this arrangement would require Mi1 and Tm3 on the outer arms of the model, which is in stark contrast to their reported 1° spatial offset [4].

Among the more direction-selective detectors was also one combination with Tm3 on the central and Mi1 on the suppressing arm. However, the resulting BL subunit considered alone shows very poor directional tuning, and the direction selectivity arises

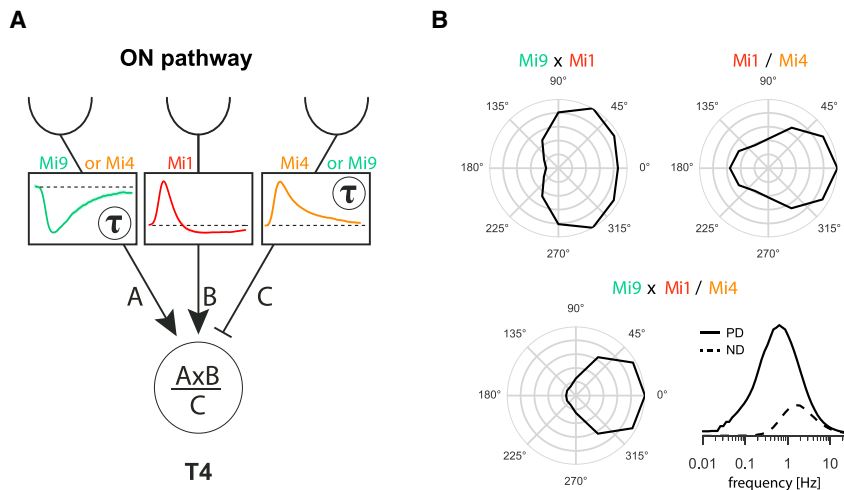


Figure 6. Proposed Implementation of the Elementary Motion Detectors in the ON Pathway

(A) T4 neurons implement both PD enhancement and ND suppression with receptive fields offset in this order along the PD axis. This requires one central fast arm being flanked by two delayed or stronger low-pass-filtered inputs. The relatively fast kinetics of Mi1 or Tm3 would suggest either or both for the central input. Mi4 and Mi9, on the other hand, show pure low-pass characteristics in their temporal kernels fitting the requirements of the two outer arms. The signs of both outer-arm synapses depend on the arrangement of Mi4 and Mi9 to accommodate their respective response polarity and match them to the required enhancing and suppressive inputs.

(B) Simulated detector responses for gratings moving across the visual field in 12 different directions, separated from each other by 30°. Top: directional tuning for the two sub-modules of this

detector. Top left: the pure HR (half) detector Mi9xMi1 shows some direction selectivity but has a low tuning sharpness. Top right: the pure BL detector Mi1/Mi4 shows a substantial response in the ND direction (180°). Bottom: directional tuning for the hybrid detector Mi9xMi1/Mi4. This hybrid detector is very sharply tuned to rightward motion (left), whereas its direction selectivity remains high across stimulus frequencies (right).

virtually exclusively from the HR subunit. Even taking the reported small anatomical offset of about 1° between these cells into account [4] did not compensate for the small temporal differences but actually resulted in even worse directional tuning. This indicates that sizable differences not only in the temporal but also in the spatial domain are a prerequisite for direction selectivity. Similar considerations are true for all combinations that place neurons with similar response kinetics on neighboring arms. In general, the most effective hybrid detectors result from combinations of cells that are arranged such that the respective HR and BL sub-detectors are as direction selective as possible and aligned in their PD.

For detectors where two low-pass filters flank a central band-pass filter element, both Mi1 and Tm3 seem feasible to fill the role of the latter. However, a previous study blocking the synaptic output of Tm3 found an effect on the response of tangential cells to moving ON edges only at high but not at low to moderate velocities [21]. Hence, although we do not exclude a functional role for Tm3 in ON motion detection, this finding argues against Tm3 as the (sole) central arm of the detector in the ON pathway, as the interference especially with the central arm should fully abolish the detection of motion.

Taken together, an implementation of the ON elementary motion detector as depicted in Figure 6 seems most likely: Mi1 as the fast central input, flanked by the low-pass elements Mi4 and Mi9 constituting the suppressing and enhancing arm in either order. Depending on the location, these neurons need to be either both excitatory or both inhibitory to accommodate their respective response polarity and fulfill the required role of enhancing and suppressing input. Considering their opposite polarity and reciprocal connection, it is more likely that both neurons are inhibitory. This would place Mi9 on the enhancing arm (“A” in Figure 6A), and Mi4 on the suppressing arm (“C” in Figure 6A). Importantly, with the observed range of temporal response characteristics in the input elements, it is not necessary to postulate further delays at the synaptic or dendritic level.

In the OFF pathway, the algorithmic structure of motion detection is less clear. On the one hand, spatiotemporal receptive field measurements of T5 neurons reveal excitatory and inhibitory sub-fields that are offset along the PD axis and appropriately tilted in space and time to support PD enhancement and ND suppression [23]. This would suggest a similar architecture as for T4. On the other hand, other studies have only reported PD enhancement for T5 [22, 39]. Nevertheless, we performed analogous simulations based on the measured T5 input kinetics and receptive fields assuming a similar detector architecture. The two detectors with the highest direction selectivity incorporated the low-pass filter, Tm9, into their suppressing arm (Figure 5E). Lacking a second pure low-pass filter input in the OFF pathway, the central and enhancing arms were occupied with band-pass filters. Because the PDs of PD enhancement and ND suppression need to be aligned, the fastest element of the combination, principally Tm2, must be located in the central position. This is also illustrated by the two worst combinations (Figure 5E, right), where even though the BL module on its own performs quite well, the oppositely oriented HR module destroys the direction selectivity of the hybrid detector. As above, hybrid detectors with the low-pass filter, Tm9, on the central arm perform poorly, as the constituting BL half contributes little to direction selectivity in those combinations (Figure 5E, right; e.g., Tm2xTm9/Tm4).

According to our simulations, and if the structure for T5 resembles the hybrid detector proposed for T4, the arrangement of a central Tm2, flanked on the null side by an enhancing Tm1 and on the preferred side by a suppressing Tm9 input, achieves by far the best direction selectivity. This implementation would predict inhibitory/suppressing input from Tm9 onto T5, which could be experimentally tested. Consistent with this arrangement, out of all four T5 columnar inputs, blocking the synaptic output from Tm4 cells results in the lowest reduction in OFF-edge responses in tangential cells [20]. Nevertheless, those blocking experiments indicate that Tm4 plays a role in the detector that awaits resolving.

In this study, we have shown that the activation of the octopamine system by CDM shifts the temporal-frequency and velocity tuning of T4 and T5 neurons to higher temporal frequencies/velocities, mirroring the temporal tuning shift in tangential cells of the lobula plate observed under active locomotion or octopaminergic activation [24–27]. At the level of T4 and T5 neurons, we find a temporal-frequency optimum of about 1–1.5 Hz for moving gratings under control conditions, corroborating previous studies [14, 17, 22]. Application of CDM shifts the temporal-frequency optimum to 2.5 Hz. T4 and T5 cells show a velocity optimum for moving edges at 3°/s or lower under control conditions, which shifts to about 12°/s under CDM. The much higher velocity optimum observed in tangential cells [40] results from the summation of synaptic inputs from the larger number of T4 and T5 neurons swept by the edge during the same time interval at higher velocities.

In parallel to the temporal-frequency tuning shift in T4 and T5 neurons, the temporal response properties of the input elements, in particular of the band-pass filter elements, accelerate. Indeed, the shift in the tuning of T4 and T5 neurons (Figure 3) can be fully accounted for by the speeding of their input elements (Figures 4 and 5). This further supports the hypothesis that the temporal kinetics of the input elements alone, without any further filtering at the synaptic or T4/T5 dendritic levels, represent the delay stage of the elementary motion detectors.

Interestingly, we observe that whereas the order of input elements with respect to their filter characteristics generally remained the same under CDM, Tm1 became faster than Tm4 (Figure S6). As a consequence, simulated motion detectors using combinations that relied on temporal differences between these two cell types suffered a strong reduction or complete breakdown of direction selectivity under CDM (Figure 5). Considering cell-to-cell variability and such changes under different physiological conditions, detectors relying on small differences in the dynamics of their input elements [18, 39] will not be robust.

Octopaminergic neurons broadly innervate the optic lobes, specifically the medulla, lobula, and lobula plate [26, 28, 29]. They activate during flight and are necessary and sufficient for the observed change in the temporal tuning profile of tangential cells [26]. Although the molecular and cellular mechanisms of action on the medulla neurons and T4/T5 cells, as well as the precise physiological activation of the octopamine system, are beyond the scope of this study, a few points are worth noting. Four different types of octopamine receptors exist in *Drosophila* that are all G protein-coupled receptors but act via different pathways and thus will have different effects [41, 42]. Of those four types, only the octopamine receptors Oamb and to a lesser degree Oct1 β R appear to be expressed in the optic lobes [42]. The expression pattern of these octopamine receptors is not known at the cellular level. Considering that all input elements in both the ON and the OFF pathway are accelerated in their responses, albeit to different degrees, it is entirely possible that those changes are indirect and inherited from neurons in the lamina or even the retina. For example, an accelerated response in L1 and L2, and to a smaller degree in L3, could explain the observed response changes in the medulla neurons described here. So far, octopaminergic neurons have not been shown to innervate the retina and lamina directly [26, 28, 29], yet octopamine might nevertheless directly or indirectly affect

photoreceptors or lamina neurons. For example, lamina wide-field neurons, projecting from the medulla back into the lamina and forming synaptic inputs to lamina neurons [43], are modulated by the behavioral state and octopamine signaling [44]. Although it cannot be excluded that octopamine acts at multiple levels, including on T4/T5 neurons directly, we have shown that the observed tuning shift in T4/T5 neurons can be fully accounted for by the changes in the temporal dynamics of their input elements.

Pharmacological activation, like any optogenetic or other exogenous activation of the octopamine system, is unlikely to capture all subtleties of the physiological changes during active locomotion, yet it can serve as a tool to manipulate the tuning of the visual motion detection system. At the same time, considering the match between pharmacological manipulation and physiological state changes observed at the level of lobula plate tangential cells [24–26], it is highly likely that the speeding of the filter characteristics in the medulla neurons described here is relevant under physiological conditions.

We have shown that it is possible to construct a hybrid HR/BL detector (as proposed in [17]) with the measured filters for the cellular elements for both the ON and the OFF pathway across different network states. From these, we can predict anatomical arrangements that would give rise to the observed response characteristics of the elementary motion detectors. Although we cannot rule out additional synaptic or dendritic filter mechanisms, we show that the temporal dynamics of the input elements alone are sufficient to explain the response properties of the elementary motion detectors across different tuning regimes. Future studies using the genetic toolbox of *Drosophila* to activate or block individual input neurons and studying the effects on visual responses in the T4 and T5 cells, as well as neurotransmitter and receptor expression pattern analyses and electron-microscopic reconstructions of the wiring, will be required to verify and further confine the proposed circuitry.

EXPERIMENTAL PROCEDURES

Experimental procedures are described in detail in the [Supplemental Experimental Procedures](#).

SUPPLEMENTAL INFORMATION

Supplemental Information includes Supplemental Experimental Procedures, six figures, and two tables and can be found with this article online at <http://dx.doi.org/10.1016/j.cub.2017.01.051>.

AUTHOR CONTRIBUTIONS

A.A., M.S.D., and A.B. conceived the study and designed the experiments. A.A. conducted and analyzed the measurements of T4/T5 cell responses. M.S.D. designed the projector-based stimulation arena and performed and analyzed the measurements of the OFF-pathway elements. F.G.R. performed and analyzed the experiments describing the ON-pathway neurons. G.A. performed and analyzed the patch-clamp recordings from lobula plate tangential cells. M.S.D. performed the computer simulations. A.A. wrote the manuscript with the help of all authors.

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REFERENCES

- Borst, A. (2014). Fly visual course control: behaviour, algorithms and circuits. *Nat. Rev. Neurosci.* 15, 590–599.
- Fischbach, K.F., and Dittrich, A.P.M. (1989). The optic lobe of *Drosophila melanogaster*. I. A Golgi analysis of wild-type structure. *Cell Tissue Res.* 258, 441–475.
- Bausenwein, B., Dittrich, A.P., and Fischbach, K.F. (1992). The optic lobe of *Drosophila melanogaster*. II. Sorting of retinotopic pathways in the medulla. *Cell Tissue Res.* 267, 17–28.
- Takemura, S.-Y., Bharioke, A., Lu, Z., Nern, A., Vitaladevuni, S., Rivlin, P.K., Katz, W.T., Olbris, D.J., Plaza, S.M., Winston, P., et al. (2013). A visual motion detection circuit suggested by *Drosophila* connectomics. *Nature* 500, 175–181.
- Shinomiyama, K., Karuppudurai, T., Lin, T.-Y., Lu, Z., Lee, C.-H., and Meinertzhagen, I.A. (2014). Candidate neural substrates for off-edge motion detection in *Drosophila*. *Curr. Biol.* 24, 1062–1070.
- Barlow, H.B., and Levick, W.R. (1965). The mechanism of directionally selective units in rabbit's retina. *J. Physiol.* 178, 477–504.
- Hassenstein, B., and Reichardt, W. (1956). Systemtheoretische Analyse der Zeit-, Reihenfolgen- und Vorzeichenauswertung bei der Bewegungsperzeption des Rüsselkäfers *Chlorophanus*. *Z. Naturforsch. B* 11, 513–524.
- Borst, A., and Helmstaedter, M. (2015). Common circuit design in fly and mammalian motion vision. *Nat. Neurosci.* 18, 1067–1076.
- Joesch, M., Schnell, B., Raghu, S.V., Reiff, D.F., and Borst, A. (2010). ON and OFF pathways in *Drosophila* motion vision. *Nature* 468, 300–304.
- Strother, J.A., Nern, A., and Reiser, M.B. (2014). Direct observation of ON and OFF pathways in the *Drosophila* visual system. *Curr. Biol.* 24, 976–983.
- Clark, D.A., Bursztyn, L., Horowitz, M.A., Schnitzer, M.J., and Clandinin, T.R. (2011). Defining the computational structure of the motion detector in *Drosophila*. *Neuron* 70, 1165–1177.
- Eichner, H., Joesch, M., Schnell, B., Reiff, D.F., and Borst, A. (2011). Internal structure of the fly elementary motion detector. *Neuron* 70, 1155–1164.
- Joesch, M., Weber, F., Eichner, H., and Borst, A. (2013). Functional specialization of parallel motion detection circuits in the fly. *J. Neurosci.* 33, 902–905.
- Maisak, M.S., Haag, J., Ammer, G., Serbe, E., Meier, M., Leonhardt, A., Schilling, T., Bahl, A., Rubin, G.M., Nern, A., et al. (2013). A directional tuning map of *Drosophila* elementary motion detectors. *Nature* 500, 212–216.
- Mauss, A.S., Meier, M., Serbe, E., and Borst, A. (2014). Optogenetic and pharmacologic dissection of feedforward inhibition in *Drosophila* motion vision. *J. Neurosci.* 34, 2254–2263.
- Mauss, A.S., Pankova, K., Arenz, A., Nern, A., Rubin, G.M., and Borst, A. (2015). Neural circuit to integrate opposing motions in the visual field. *Cell* 162, 351–362.
- Haag, J., Arenz, A., Serbe, E., Gabbiani, F., and Borst, A. (2016). Complementary mechanisms create direction selectivity in the fly. *eLife* 5, e17421.
- Behnia, R., Clark, D.A., Carter, A.G., Clandinin, T.R., and Desplan, C. (2014). Processing properties of ON and OFF pathways for *Drosophila* motion detection. *Nature* 512, 427–430.
- Behnia, R., and Desplan, C. (2015). Visual circuits in flies: beginning to see the whole picture. *Curr. Opin. Neurobiol.* 34, 125–132.
- Serbe, E., Meier, M., Leonhardt, A., and Borst, A. (2016). Comprehensive characterization of the major presynaptic elements to the *Drosophila* OFF motion detector. *Neuron* 89, 829–841.
- Ammer, G., Leonhardt, A., Bahl, A., Dickson, B.J., and Borst, A. (2015). Functional specialization of neural input elements to the *Drosophila* ON motion detector. *Curr. Biol.* 25, 2247–2253.
- Fisher, Y.E., Silies, M., and Clandinin, T.R. (2015). Orientation selectivity sharpens motion detection in *Drosophila*. *Neuron* 88, 390–402.
- Leong, J.C.S., Esch, J.J., Poole, B., Ganguli, S., and Clandinin, T.R. (2016). Direction selectivity in *Drosophila* emerges from preferred-direction enhancement and null-direction suppression. *J. Neurosci.* 36, 8078–8092.
- Chiappe, M.E., Seelig, J.D., Reiser, M.B., and Jayaraman, V. (2010). Walking modulates speed sensitivity in *Drosophila* motion vision. *Curr. Biol.* 20, 1470–1475.
- Jung, S.N., Borst, A., and Haag, J. (2011). Flight activity alters velocity tuning of fly motion-sensitive neurons. *J. Neurosci.* 31, 9231–9237.
- Suver, M.P., Mamiya, A., and Dickinson, M.H. (2012). Octopamine neurons mediate flight-induced modulation of visual processing in *Drosophila*. *Curr. Biol.* 22, 2294–2302.
- Longden, K.D., and Krapp, H.G. (2010). Octopaminergic modulation of temporal frequency coding in an identified optic flow-processing interneuron. *Front. Syst. Neurosci.* 4, 153.
- Sinakevitch, I., and Strausfeld, N.J. (2006). Comparison of octopamine-like immunoreactivity in the brains of the fruit fly and blow fly. *J. Comp. Neurol.* 494, 460–475.
- Busch, S., Selcho, M., Ito, K., and Tanimoto, H. (2009). A map of octopaminergic neurons in the *Drosophila* brain. *J. Comp. Neurol.* 513, 643–667.
- Chen, T.-W., Wardill, T.J., Sun, Y., Pulver, S.R., Renninger, S.L., Baohan, A., Schreiter, E.R., Kerr, R.A., Orger, M.B., Jayaraman, V., et al. (2013). Ultrasensitive fluorescent proteins for imaging neuronal activity. *Nature* 499, 295–300.
- Roeder, T. (1995). Pharmacology of the octopamine receptor from locust central nervous tissue (OAR3). *Br. J. Pharmacol.* 114, 210–216.
- Schnell, B., Weir, P.T., Roth, E., Fairhall, A.L., and Dickinson, M.H. (2014). Cellular mechanisms for integral feedback in visually guided behavior. *Proc. Natl. Acad. Sci. USA* 111, 5700–5705.
- Mazurek, M., Kager, M., and Van Hooser, S.D. (2014). Robust quantification of orientation selectivity and direction selectivity. *Front. Neural Circuits* 8, 92.
- Meier, M., Serbe, E., Maisak, M.S., Haag, J., Dickson, B.J., and Borst, A. (2014). Neural circuit components of the *Drosophila* OFF motion vision pathway. *Curr. Biol.* 24, 385–392.
- Fisher, Y.E., Leong, J.C.S., Sporar, K., Ketkar, M.D., Gohl, D.M., Clandinin, T.R., and Silies, M. (2015). A class of visual neurons with wide-field properties is required for local motion detection. *Curr. Biol.* 25, 3178–3189.
- Yang, H.H., St-Pierre, F., Sun, X., Ding, X., Lin, M.Z., and Clandinin, T.R. (2016). Subcellular imaging of voltage and calcium signals reveals neural processing in vivo. *Cell* 166, 245–257.
- Takemura, S.-Y., Xu, C.S., Lu, Z., Rivlin, P.K., Parag, T., Olbris, D.J., Plaza, S., Zhao, T., Katz, W.T., Umayam, L., et al. (2015). Synaptic circuits and their variations within different columns in the visual system of *Drosophila*. *Proc. Natl. Acad. Sci. USA* 112, 13711–13716.
- Silies, M., Gohl, D.M., Fisher, Y.E., Freifeld, L., Clark, D.A., and Clandinin, T.R. (2013). Modular use of peripheral input channels tunes motion-detecting circuitry. *Neuron* 79, 111–127.

39. Salazar-Gatzimas, E., Chen, J., Creamer, M.S., Mano, O., Mandel, H.B., Matulis, C.A., Pottackal, J., and Clark, D.A. (2016). Direct measurement of correlation responses in *Drosophila* elementary motion detectors reveals fast timescale tuning. *Neuron* 92, 227–239.
40. Leonhardt, A., Ammer, G., Meier, M., Serbe, E., Bahl, A., and Borst, A. (2016). Asymmetry of *Drosophila* ON and OFF motion detectors enhances real-world velocity estimation. *Nat. Neurosci.* 19, 706–715.
41. Farooqui, T. (2007). Octopamine-mediated neuromodulation of insect senses. *Neurochem. Res.* 32, 1511–1529.
42. El-Kholy, S., Stephano, F., Li, Y., Bhandari, A., Fink, C., and Roeder, T. (2015). Expression analysis of octopamine and tyramine receptors in *Drosophila*. *Cell Tissue Res.* 361, 669–684.
43. Rivera-Alba, M., Vitaladevuni, S.N., Mishchenko, Y., Lu, Z., Takemura, S.Y., Scheffer, L., Meinertzhagen, I.A., Chklovskii, D.B., and de Polavieja, G.G. (2011). Wiring economy and volume exclusion determine neuronal placement in the *Drosophila* brain. *Curr. Biol.* 21, 2000–2005.
44. Tuthill, J.C., Nem, A., Rubin, G.M., and Reiser, M.B. (2014). Wide-field feedback neurons dynamically tune early visual processing. *Neuron* 82, 887–895.

Current Biology, Volume 27

Supplemental Information

**The Temporal Tuning of the *Drosophila*
Motion Detectors Is Determined
by the Dynamics of Their Input Elements**

Alexander Arenz, Michael S. Drews, Florian G. Richter, Georg Ammer, and Alexander Borst

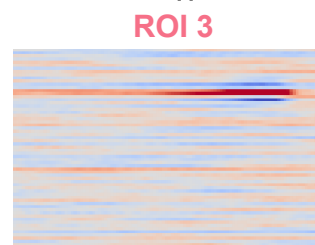
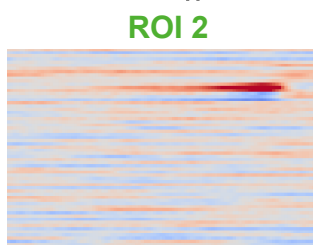
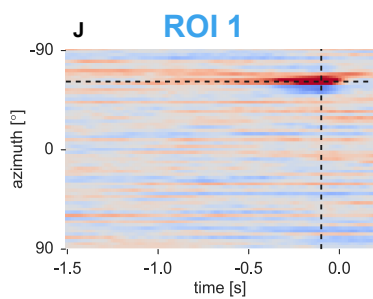
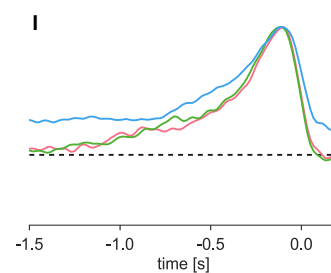
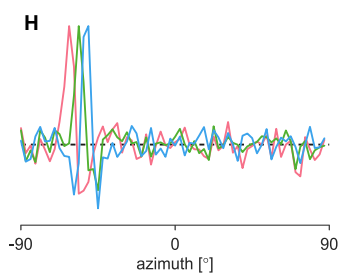
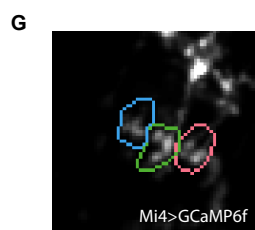
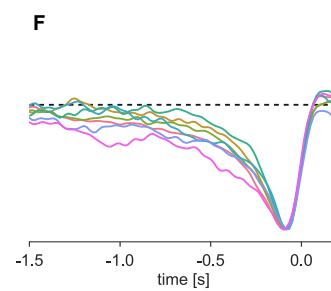
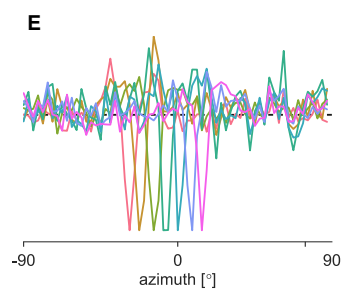
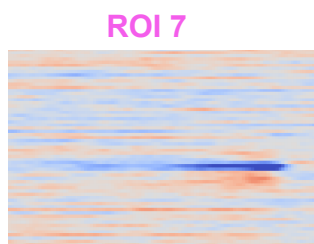
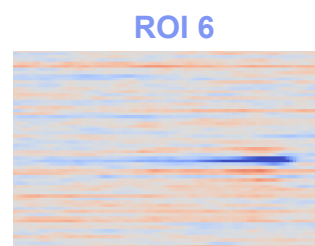
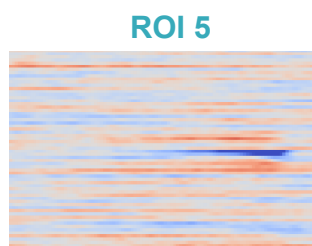
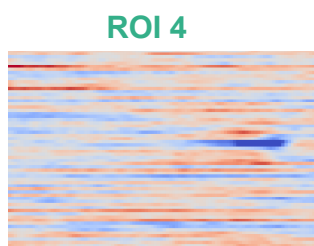
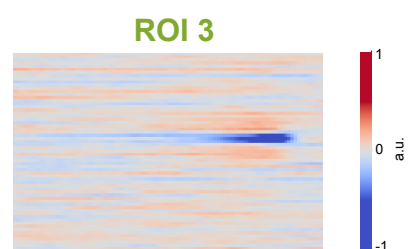
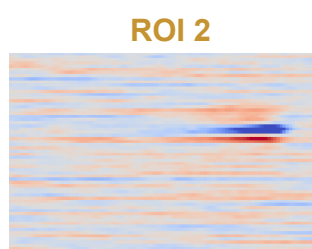
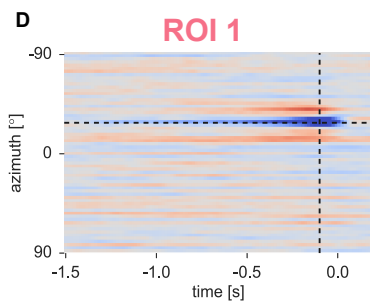
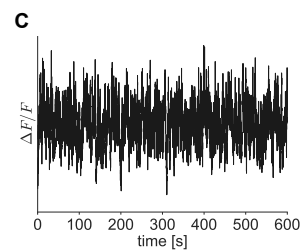
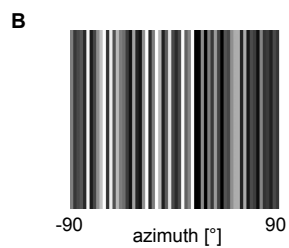
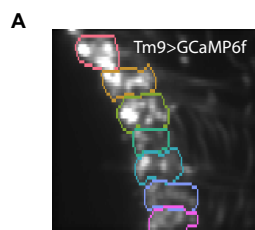


Figure S1. Related to Figure 2. Example acquisition of spatiotemporal receptive fields via stochastic stimulation and reverse correlation of calcium signals for the neurons Tm9 and Mi4.

(A) 2-photon image from a fly expressing GCaMP6f in Tm9 axon terminals in the lobula. Highlighted in color are seven manually drawn regions of interest (ROIs) around individual terminals from neighboring columns.

(B) Snapshot of one frame of the one-dimensional horizontal noise stimulus.

(C) Calcium trace from a single ROI in response to 10 minutes of white noise stimulation.

(D) Spatiotemporal receptive fields obtained by reverse correlation of the calcium signals in each ROI with the stimulus.

(E) Cross-sections through the receptive fields along the space axis reflecting the retinotopic organisation of the lobula.

(F) Cross-sections through the receptive fields along the temporal axis revealing the low-pass characteristics of Tm9.

(G-J) Same for Mi4.

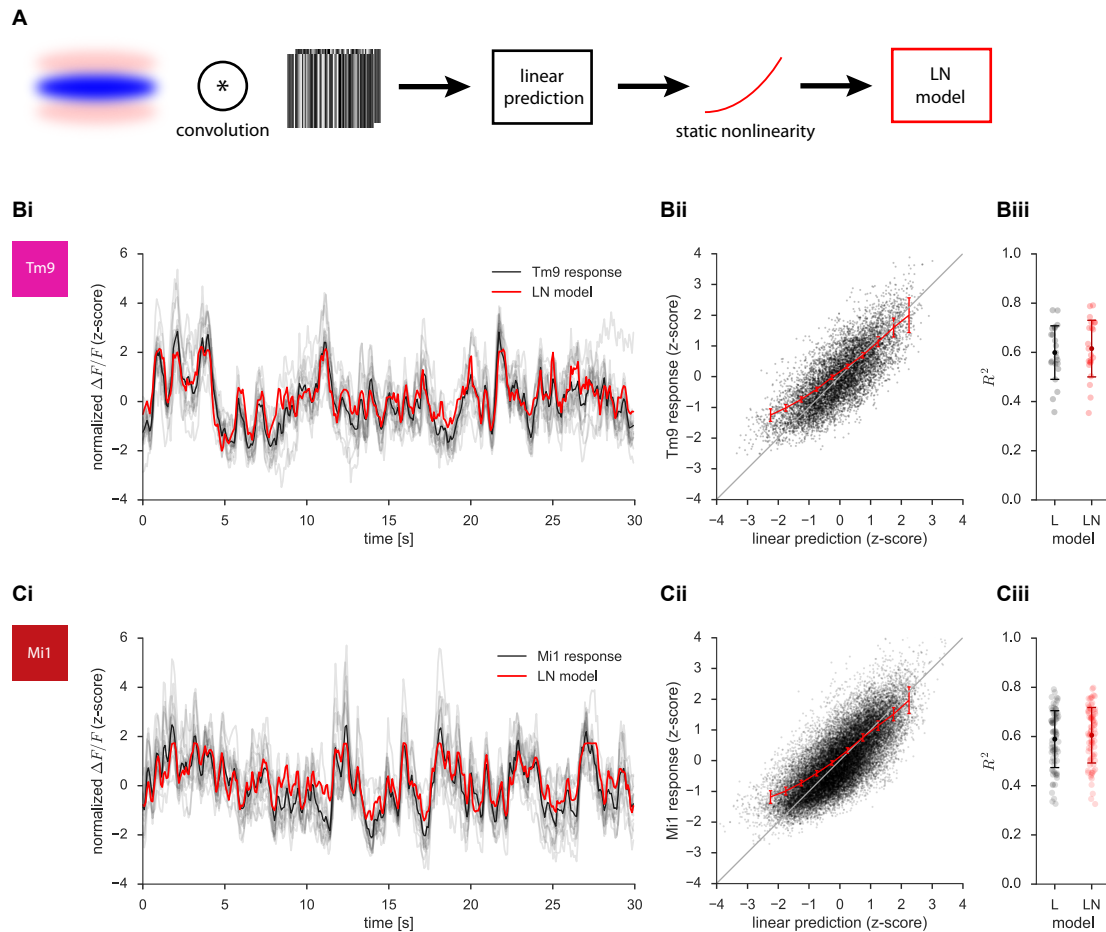


Figure S2. Related to Figure 2. Prediction of calcium responses in Tm9 and Mi1 terminals from the linear spatiotemporal receptive fields.

(A) Schematic of the model. The linear prediction of individual axon terminal responses (of Tm9 or Mi1) to a white noise stimulus is given by the convolution of the stimulus with the respective spatio-temporal receptive field of the cell. A linear-nonlinear model (LN) is built by remapping the output of the linear prediction with a static nonlinearity.

(Bi) Actual response of an exemplary Tm9 axon terminal (black) and the prediction of the LN model (red). (Bii) Scatter plot of the linear prediction against the actual response for all cells recorded. The static nonlinearity (red) is obtained by averaging the point cloud within discrete bins along the x-axis for each axon terminal. (Biii) Coefficient of determination for the linear model (L, black) and the linear-nonlinear model (LN, red). The linear model prediction alone accounted for 60% and the LN model for 62% of the response variance. Circles represent measurements of individual terminals, the bar shows the standard deviation and the mean among all cells measured ($N = 4$, $n = 22$).

(C) Same as in (B), but for Mi1 ($N = 4$, $n = 78$). The L model alone accounted for 59% and the LN model for 61% of the response variance.

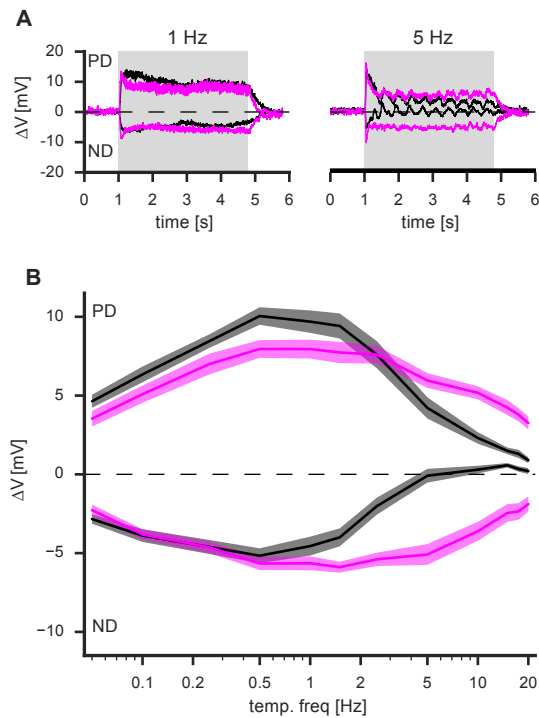
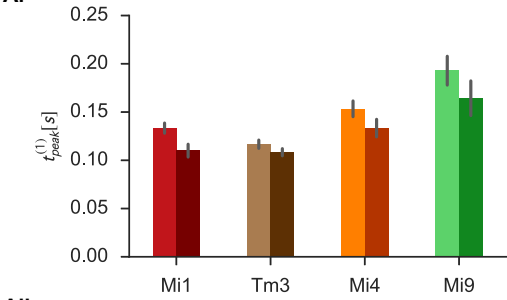


Figure S3. Related to Figure 3. Temporal frequency tunings of lobula plate tangential cells change with the application of the octopamine agonist CDM.

(A) Voltage responses of HS and VS tangential cells in the lobula plate (population average, $N = 15$ flies, $n=15$ cells) to square-wave gratings moving in the preferred or null direction in control (black) and after application of CDM (magenta) for gratings moving at a temporal frequency of 1 Hz (left) or 5 Hz (right). The period of motion of the grating is indicated by the grey-shaded region.

(B) Average voltage responses over the stimulation period for square-wave gratings at different temporal frequencies. Responses, measured as average voltage deflections over the whole stimulus period, peaked at 0.5 Hz in both the preferred (as maximum average depolarization) and null (as maximum average hyperpolarization) direction. Application of the octopamine agonist chlordimeform (CDM; magenta) at a final concentration of 20 μM resulted in increased responses to higher temporal frequencies from 2-20 Hz.

Ai

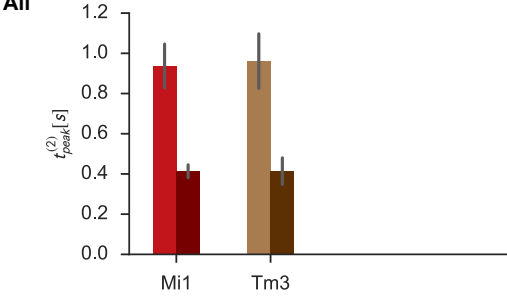


Aiv

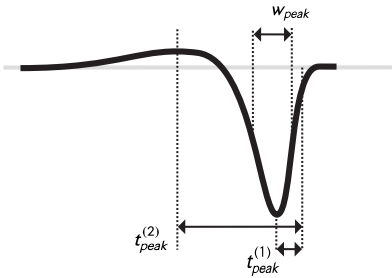
	$t_{peak}^{(1)}$	$t_{peak}^{(2)}$	w_{peak}
Mi1	0.108	0.013	0.014
Tm3	0.076	0.004	< 0.001
Mi4	0.033	n.a	0.041
Mi9	0.259	n.a	0.037

control vs CDM

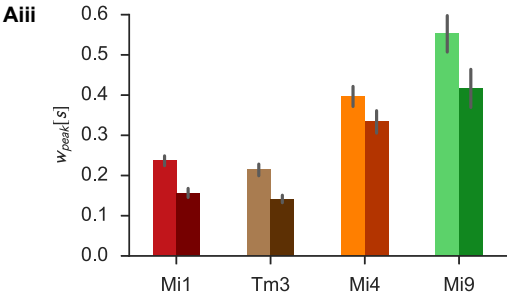
Aii



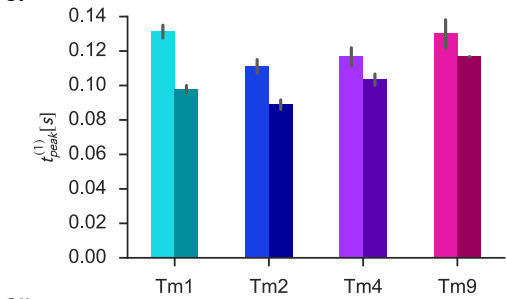
B



Aiii



Ci

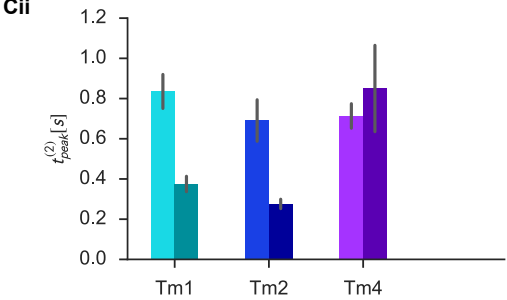


Civ

	$t_{peak}^{(1)}$	$t_{peak}^{(2)}$	w_{peak} [s]
Tm1	< 0.001	0.004	< 0.001
Tm2	< 0.001	0.005	< 0.001
Tm4	0.016	0.564	0.003
Tm9	0.178	n.a	0.015

control vs CDM

Cii



Ciii

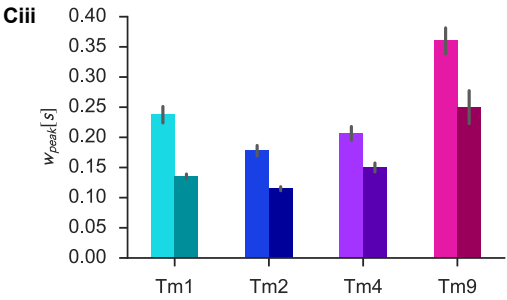


Figure S4. Related to Figure 2 and 4. Analysis of the temporal filters of the inputs to T4 and T5.

(A) T4 inputs. We quantified the shape of the temporal filters, as well as their change after application of CDM, by three measures (B): the time-to-peak for the first peak ($t_{\text{peak}}^{(1)}$, (Ai)) and, for the biphasic filter kernels of band-pass filters, to the second peak ($t_{\text{peak}}^{(2)}$, (Aii)), as well as the full-width at half-maximum for the first peak (w_{peak} , (Aiii)) of the temporal kernel. Measurements after application of CDM are presented in a darker color shade (right bars) than for the control condition (left bars) of the respective cell. Statistical comparisons between control and CDM condition (based on a paired t-test) are shown in (Aiv), highlighted in red color when statistical significance is observed (with a gradient in the red nuance from light to dark red indicating p values of $p < 0.05$, $p < 0.01$ and $p < 0.001$).

(C) Same, for T5 inputs.

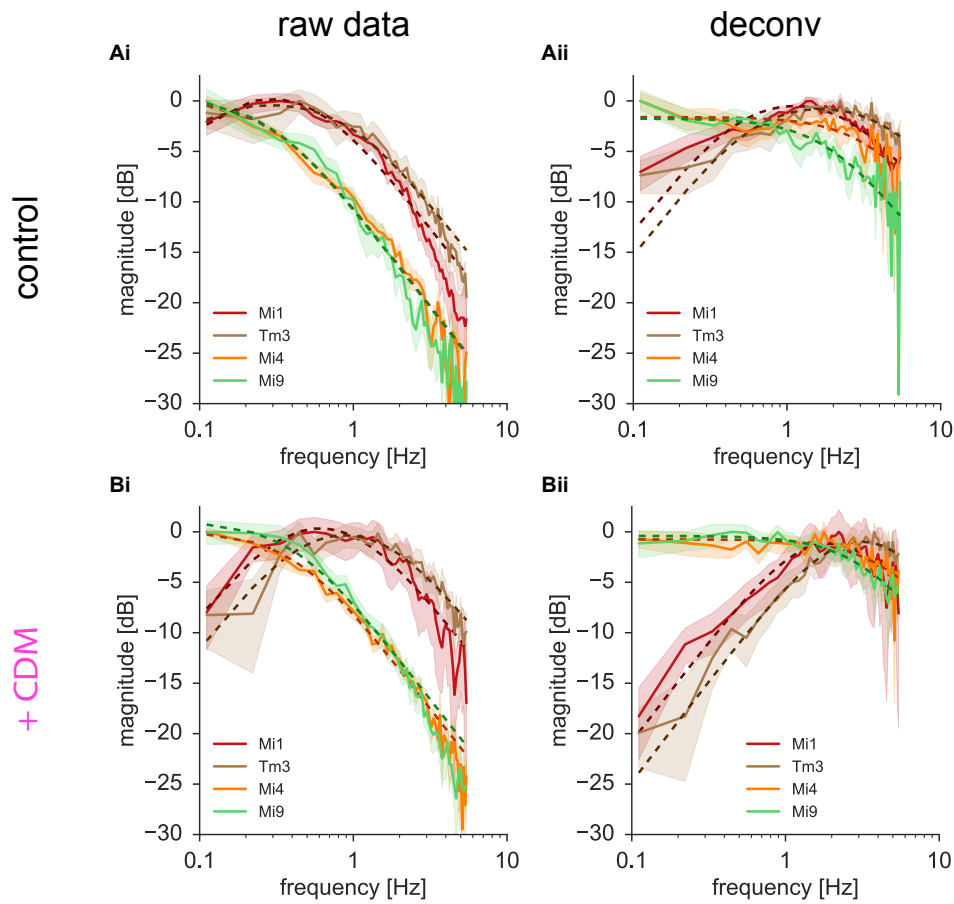


Figure S5. Related to Figure 5. Model fitting on the frequency spectra of the ON pathway elements.

(Ai) Frequency spectrum derived from calcium imaging experiments for the ON pathway columnar neurons Mi1, Tm3, Mi4, Mi9. (Aii) Frequency spectrum after deconvolution with a low-pass filter representing the dynamics of the calcium indicator GCaMP6f. Dashed lines represent the fitted frequency responses of 1st order band-pass or low-pass filters.

(Bi, Bii) Like (Ai, Aii), for the spectra determined from the recordings after application of CDM.

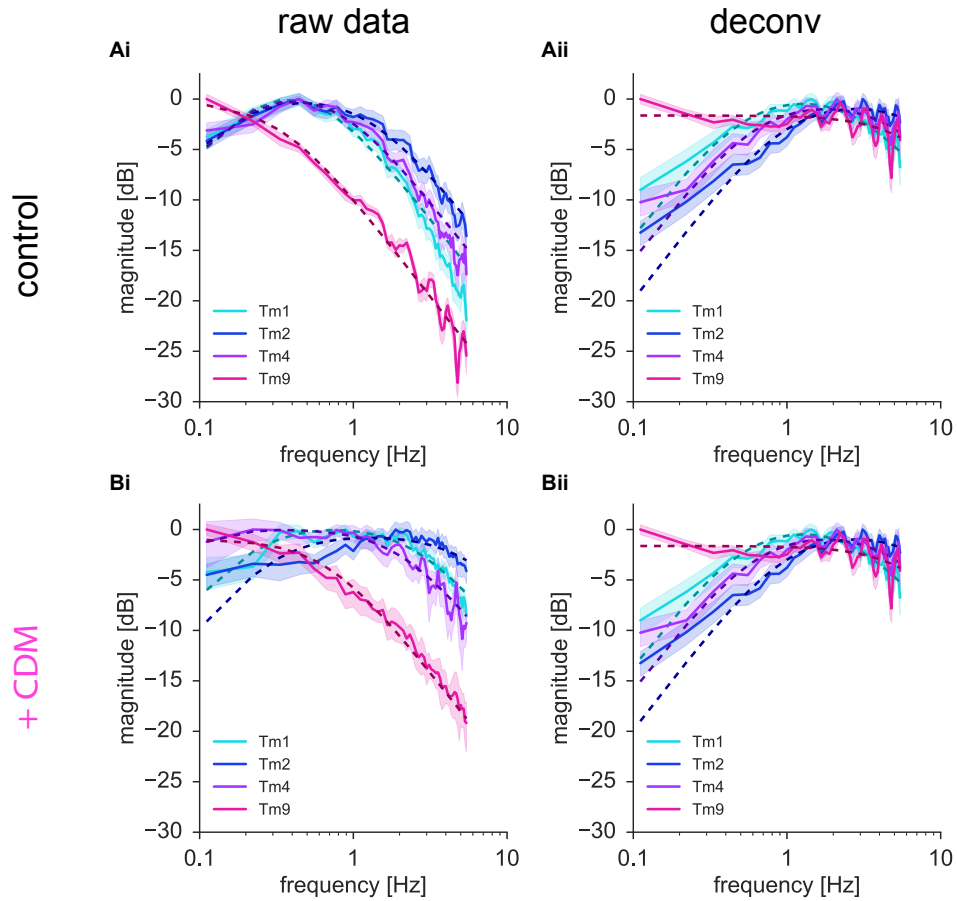


Figure S6. Related to Figure 5. Model fitting on the temporal filter frequency spectra of the OFF pathway elements.

(Ai) Measured frequency spectra based on calcium imaging experiments for the OFF pathway elements Tm1, Tm2, Tm4 and Tm9. (Aii) Frequency spectra after deconvolution with a filter describing the dynamics of the calcium indicator. Dashed lines represent the fitted frequency responses of 1st order band-pass or low-pass filters.

(Bi, Bii) Same as (Ai, Aii), after application of CDM.

	Mi1		Tm3		Mi4		Mi9	
	control	+ CDM	control	+ CDM	control	+ CDM	control	+ CDM
A_{raw}	1.271	1.314	0.996	1.204	1.007	1.000	0.982	1.060
$t_{\text{HP, raw}}$	1.078 s	0.445 s	1.769 s	0.340 s				
$t_{\text{LP, raw}}$	0.266 s	0.143 s	0.158 s	0.086 s	0.519 s	0.370 s	0.546 s	0.373 s
$R^2_{\text{t, raw}}$	0.975	0.942	0.971	0.925	0.984	0.990	0.982	0.987
A_{deconv}	1.146	1.643	1.035	2.034	0.831	0.906	0.789	0.940
$t_{\text{HP, deconv}}$	0.318 s	0.075 s	0.260 s	0.044 s				
$t_{\text{LP, deconv}}$	0.054 s	0.075 s	0.027 s	0.044 s	0.038 s	0.028 s	0.077 s	0.043 s
$R^2_{\text{t, deconv}}$	0.872	0.851	0.768	0.955	0.666	0.643	0.832	0.906
$A_{\text{sur}} / A_{\text{cen}}$	0.022	0.028	0.000	0.000	0.132	0.093	0.063	0.043
FWHM_{cen}	6.81 °	7.09 °	11.91 °	12.40 °	6.47 °	6.49 °	6.37 °	6.37 °
FWHM_{sur}	28.81 °	19.86 °	-	-	16.14 °	20.02 °	23.98 °	28.08 °
R^2_s	0.995	0.990	0.985	0.984	0.987	0.989	0.980	0.992

Table S1. Related to Figure 2 and 4. Spatio-temporal response properties of T4 input cells.

Numerical parameters derived from the model fits to the temporal and the spatial components of the spatio-temporal receptive fields obtained from reverse correlation for the ON-pathway neurons.

	Tm1		Tm2		Tm4		Tm9	
	control	+ CDM	control	+ CDM	control	+ CDM	control	+ CDM
A_{raw}	1.419	1.072	1.068	0.933	1.218	0.990	0.961	0.890
$t_{\text{HP, raw}}$	0.632 s	0.754 s	0.962 s	0.558 s	0.788 s	2.210 s		
$t_{\text{LP, raw}}$	0.271 s	0.051 s	0.113 s	0.020 s	0.186 s	0.070 s	0.462 s	0.220 s
$R^2_{\text{t, raw}}$	0.978	0.901	0.979	0.564	0.972	0.930	0.985	0.981
A_{deconv}	1.117	2.217	1.038	2.657	1.018	1.002	0.827	0.836
$t_{\text{HP, deconv}}$	0.296 s	0.032 s	0.153 s	0.018 s	0.249 s	0.086 s		
$t_{\text{LP, deconv}}$	0.044 s	0.032 s	0.014 s	0.018 s	0.024 s	0.015 s	0.017 s	0.008 s
$R^2_{\text{t, deconv}}$	0.895	0.974	0.893	0.990	0.807	0.880	0.273	0.226
$A_{\text{sur}} / A_{\text{cen}}$	0.040	0.029	0.035	0.026	0.054	0.005	0.046	0.071
FWHM_{cen}	8.12 °	7.34 °	7.93 °	7.23 °	11.45 °	10.10 °	6.92 °	6.53 °
FWHM_{sur}	27.14 °	20.46 °	30.52 °	23.01 °	34.62 °	72.03 °	23.78 °	21.74 °
R^2_{s}	0.997	0.995	0.994	0.991	0.992	0.988	0.995	0.995

Table S2. Related to Figure 2 and 4. Spatio-temporal response properties of T5 input cells.

Numerical parameters derived from the model fits to the temporal and the spatial components of the spatio-temporal receptive fields obtained from reverse correlation for the OFF-pathway neurons.

Supplemental Experimental Procedures

Flies/preparation

Flies were raised and kept on standard cornmeal-agar medium on a 12 hour light/12 hour dark cycle at 25°C and 60% humidity. For patch-clamp recordings from tangential cells, Canton S flies were used. For calcium imaging experiments, the genetically-encoded calcium indicators GCaMP6f or GCaMP6m [S1] were expressed using the Gal4/UAS- or LexA/lexAop-system in cell-type specific driver lines, resulting in the following genotypes:

Short name	Genotype
Mi1>GCa6f	<i>w⁻; R19F01-AD/UAS-GCaMP6f; R71D01-DBD/UAS-GCaMP6f</i>
Tm3>GCa6f	<i>w⁻; UAS-GCaMP6f; R13E12-Gal4</i>
Mi4>GCa6f	<i>w⁻; R48A07-AD/UAS-GCaMP6f; R13F11-DBD/UAS-GCaMP6f</i>
Mi9>GCa6f	<i>w⁻; R48A07-AD/UAS-GCaMP6f; VT046779-DBD/UAS-GCaMP6f</i>
Tm1>GCa6f	<i>w⁻; UAS-GCaMP6f; VT12717-Gal4</i>
Tm2>GCa6f	<i>w⁻; UAS-GCaMP6f; VT12282-Gal4</i>
Tm4>GCa6f	<i>w⁻; UAS-GCaMP6f; R35H01-Gal4</i>
Tm9>GCa6f	<i>w⁻; UAS-GCaMP6f; VT65303-Gal4</i>
T4/T5>GCa6m	<i>w⁻; Sp/CyO ; VT50384-lexA, lexAop-GCaMP6m/TM6b</i>

The transgenic fly lines driving split-Gal4 expression in the medulla neurons Mi1, Mi4 and Mi9, respectively, were generated and will be described in [S2] (with the Mi1 driver line corresponding to their transgenic fly line SS00809, Mi4 to SS01019, and Mi9 to SS02432).

For electrophysiological and calcium imaging experiments, flies were prepared as previously described [S3, S4]. Briefly, flies were anaesthetized on ice or with CO₂, fixed with their backs, legs and wings to a Plexiglas holder with the back of the head exposed to a recording chamber filled with fly external solution. The cuticula at the back of the head on one side was cut away with a fine hypodermic needle and removed together with muscles and air sacks covering the underlying optic lobe. To gain access to tangential cells for electrophysiological recordings, the neurolemma covering the brain was partially digested by applying 0.5mg/ml collagenase IV (Gibco) with a glass electrode to the brain until the tangential cell somata were exposed. Where indicated, the octopamine agonist chlordimeform (CDM, Sigma Aldrich) was added as a 2mM stock solution (in external solution) directly to the bath to yield a final concentration of 20 μ M. Diffusion was allowed for 15 min before recordings recommenced.

Patch-clamp recordings from vertical and horizontal system tangential cells were performed as previously described [S4].

2-Photon calcium imaging

Calcium imaging was performed on custom-built 2-photon microscopes as previously described [S3] controlled with the ScanImage software in Matlab [S5]. Acquisition rates were between 3.8 and 15 Hz, image resolution between 64x64 and 128x128 pixels. Before starting the acquisition, we verified that the receptive fields of the cells were located on the stimulus arena by showing a search stimulus consisting of moving gratings.

Visual stimulation

For the study of visual responses of lobula plate tangential cells and T4/T5 neurons, visual stimuli were presented on an **LED arena**, based on a design by [S6], covering approximately 180° in azimuth and 90° in elevation. Stimuli covered the whole extent of the arena and were presented at full contrast. Square-wave gratings had a spatial wavelength of 24°, and moved with velocities of 1.2-480°/s in the preferred and null direction, corresponding to temporal frequencies ranging from 0.05 to 20°/s. Single stimulation periods of moving gratings lasted for 3.8 s, separated by periods of 5 s where the grating remained stationary. For the edge velocity tuning, bright or dark edges of full contrast were presented, moving at velocities of 3 to 300 °/s in the preferred (up) and null direction (down) of T4c/T5c neurons, separated by 6 s. All stimuli were presented in a pseudo-random order with 3-5 repetitions per stimulus.

The spatio-temporal response properties of the Mi and Tm columnar input elements were determined on a custom-built **projector-based arena** that allowed for greater stimulus flexibility. Stimuli were projected with 2 commercial micro-projectors (TI DLP Lightcrafter 3000) onto the back of an opaque cylindrical screen covering 180° in azimuth and 105° in elevation of the fly's visual field. The projectors were programmed to use only the green LED (OSRAM L CG H9RN) which emits light between 500nm to 600nm wavelength. This increased the refresh rate from 60 to 180 Hz (at 8 bit color depth). To prevent overlap between the spectra of the GCaMP signal and the arena light, we placed two long-pass filters (Thorlabs FEL0550 and FGL550) in front of each projector restricting the stimulus light to wavelengths above 550nm. A band-pass filter in front of the photomultiplier (Brightline 520/35) allowed only the portion of the light within the GCaMP emission spectrum to be detected. Additional shielding of stray light from the arena with black foil effectively suppressed any leak of the arena light into the photomultiplier signal. The maximum luminance achieved by our stimulation system is $276 \pm 48 \text{ cd/m}^2$. For all stimuli used here, we set the medium brightness to a 8-bit grayscale value of 50, which corresponds to a medium luminance of $55 \pm 11 \text{ cd/m}^2$.

Stimuli were rendered using a custom written software in Python 2.7. To account for the curvature of the arena screen, our software pre-distorts the generated images such that the projected image appears as a regular grating on the screen. For that, the software takes advantage of functions from Panda3D, a framework for 3D rendering for Python.

Gaussian noise stimulus

To generate the horizontal white noise stimulus, we partitioned the cylindrical screen into 64 bars, so that each bar covered an angle of approximately 2.8° in azimuth. For each bar, samples were drawn at a frame rate of 60 Hz from a Gaussian distribution, so that the standard deviation was at 25% contrast around a mean intensity value of 50 on the 8-bit grayscale of the display devices. We then filtered the random samples for each bar with a Gaussian filter with a standard deviation of 5 Hz in the frequency domain which leads to a stimulus auto-correlation function that is a Gaussian with approximately 45ms standard deviation. Since the calcium indicator dynamics of GCaMP and the data acquisition frame rate (12 Hz in this case) place a lower bound on the temporal precision of the signal we can extract from calcium imaging experiments, we restricted the frequency content of the stimulus in this way to the relevant domain. The whole stimulus sequence was 10 minutes long and was exported as a video file in H.264 format with lossless compression. For the vertical noise the same stimulus was rotated by 90° and scaled such that 54 bars covered the height of the screen, accounting for the aspect ratio of the screen being approximately 1.2.

Data acquisition and analysis

Data analysis was performed offline using custom-written routines in Matlab and Python 2.7 (with the SciPy and OpenCV-Python Libraries).

For the **electrophysiological experiments**, baseline-subtracted voltage responses of tangential cells were averaged across trials, and the response to gratings was quantified as the average voltage over the whole period of the respective stimulus presentation. Preferred direction was front-to-back for HS and down for VS cells, null direction the corresponding opposite direction. For Suppl. Figure S3 voltage responses over the individual stimuli were averaged across all cells.

Calcium imaging: Images were automatically registered using vertical and horizontal translations to correct for the movement of the brain. Fluorescence changes ($\Delta F/F$ values) were then calculated by dividing every registered frame by the average of the registered first 5 images of the recording. Regions of interest (ROIs) were selected on the average raw image by hand: in layer 10 of the medulla for the ON, in the lobula for the OFF pathway elements, outlining single terminals. For T4 and T5 neurons, ROIs were routinely chosen in the lobula plate, encompassing small regions with single to few axon terminals, or selected to cover single neurites between medulla or lobula and lobula plate. For Figure 3D&E, ROIs were drawn in the medulla for T4 and in the lobula for T5 neurons to separate those 2 cell types. Averaging the fluorescence change over this ROI in space resulted in a $\Delta F/F$ time course. Neuronal responses were quantified as the maximum $\Delta F/F$ value over the stimulation period plus the subsequent 0.5 s, subtracted by the average of the baseline period covering the 2 frames before the respective stimulus onset. To average across cells/ROIs, responses were first normalized to the maximum response of each ROI to the corresponding stimulus set. For edges, normalization was performed separately to ON and OFF stimuli to take any selection bias for T4 or T5 cells within the ROI into account.

White noise reverse-correlation

For the input elements, spatio-temporal receptive fields were calculated following standard reverse-correlation methods (Figure S1) [S7, S8]. First, the mean value was subtracted from the raw signals of single ROIs by using a low-pass filtered version of the signal (Gaussian filter with 120 seconds standard deviation) as a baseline for a $\Delta F/F$ -like representation of the signal. This effectively removed slow baseline fluctuations caused by bleaching and very slow changes in the average calcium level from the signals.

We then calculated the stimulus-response reverse correlation function

$$K(x, \tau) = \int_0^T dt S(x, t - \tau) \cdot R(t)$$

where S denotes the stimulus and R the response of the neuron.

The resulting spatiotemporal fields were normalized in z-score and as a quality control only receptive fields with peak amplitudes above 10 standard deviations from the mean were taken for further analysis (for Mi9 the threshold was lowered to 7). Cross-sections through the receptive fields along the space axis were fit with a Gaussian function to determine the position of the peak.

Since one imaging frame is built up continuously over one sample time, ROIs lying at different y-coordinates in the image will in fact be imaged at slightly different times. Since the stimulus is presented at a higher frame rate of 60 Hz, this leads to a notable peak shift between the impulse responses of different ROIs. We corrected for this by translating the spatiotemporal receptive field of each ROI by a) the time difference between the start of a frame and the effective sampling point estimated by the y-coordinate of the center of mass of the respective ROI and b) the start time of the white noise stimulus within the very first frame acquired during stimulation.

Spatio-temporal receptive fields resulting from different ROIs (that were retinotopically shifted) were then centered about each other to generate a mean receptive field. To ensure receptive fields of input elements were fully covered, cells with a receptive field center less than 10 pixels (28°) from the edge of the arena were excluded.

Frozen noise

Filter kernels were validated by testing their ability to predict the neuronal responses from the stimulus. For this, neurons were again stimulated with a white-noise stimulus, only this time part of the stimulus consisted of 15 repetitions (each 30 seconds long) of a white-noise sequence ('frozen noise') to eliminate noise in the neuronal responses. As above, spatio-temporal filter kernels were then reconstructed from responses to single repetition stimulus sequences (20 minutes long). Analogously to above, only receptive fields with a peak higher than 20 standard deviations were included for further analysis. Subsequently the averaged response during the held-out test portion of the stimulus was predicted for each recorded cell individually. Linear predictions were obtained by convolution of the spatio-temporal filter kernels with the frozen noise stimulus along the time axis. Filter kernels were thresholded versions of the spatio-temporal receptive fields (all values below 5% of the peak amplitude as well as regions further away than 15° from the receptive field center were set to zero). Both, the predicted response trace and the actual mean response to the frozen noise stimulus, were normalized in z-score in order to make different cells with varying calcium indicator expression levels and therefore different absolute signal values comparable. The static nonlinearity for the LN model was estimated for each cell by averaging all values from the actual mean response corresponding to values of the predicted response within bins of size 0.5 from -2.5 to +2.5 z-score (see scatter plots Bii and Cii in Suppl. Figure S2). Prediction accuracy of the linear filter was assessed through the correlation of the predicted versus actual response of the neuron [S9].

Spatial receptive field model

The one-dimensional spatial receptive fields (Figure 2 E-H and K-N, top and right) are cross-sections through the peak of the spatiotemporal receptive fields along the space axis and are averaged over the 12 samples (200ms) around the peak. For almost all columnar neurons measured we found a small-field, antagonistic center-surround organization of the spatial receptive field using both the horizontal and the vertical white noise stimulus.

Mathematically, receptive fields of this kind can be described as a difference of Gaussians

$$RF_{1D}(\varphi) = e^{-\frac{\varphi^2}{2\sigma_{cen}^2}} - A_{rel} \cdot e^{-\frac{\varphi^2}{2\sigma_{sur}^2}}$$

without loss of generality for the horizontal one-dimensional receptive field along the azimuth φ . Here, σ_{cen} and σ_{sur} are the standard deviations of center and surround, respectively, and $A_{rel} = A_{sur}/A_{cen}$ denotes the relative strength of the surround in relation to the amplitude of the center Gaussian (which is normalized to 1).

To reconstruct a two-dimensional receptive field from the measured one-dimensional projections, we chose the same mathematical approach as above, only in 2D:

$$RF_{2D}(\varphi, \vartheta) = e^{-\frac{(\varphi^2 + \vartheta^2)}{2\sigma_{cen}^2}} - A_{rel} \cdot e^{-\frac{(\varphi^2 + \vartheta^2)}{2\sigma_{sur}^2}}$$

For simplicity, throughout the analysis we used the small-angle approximation $\tan \vartheta \approx \vartheta$ for the vertical axis or the elevation ϑ even if receptive fields span angles larger than 5° . Thus, we neglected perspective distortions induced by the arena screen not being spherical, but cylindrical. Accounting for additional distortions induced by the relative displacement of the fly's body in relation to the elevation of the receptive field on the arena would require even more detailed mathematical description, yet we did not observe any severe irregularities in the spatial receptive fields.

It is important to note that receptive field estimation via a one-dimensional stimulus as performed here yields in fact a projection of the underlying two-dimensional spatial receptive field:

$$RF_{1D}(\varphi) = \int_{-\infty}^{\infty} RF_{2D}(\varphi, \vartheta) d\vartheta$$

Hence, we fitted the above function $RF_{2D}(\varphi, \vartheta)$ such that its projections along the horizontal and vertical axis would agree with the given one-dimensional receptive field projections measured via reverse correlation. The fitting procedure was implemented using standard least-square algorithms (SciPy 0.16.1). The resulting values for A_{rel} , σ_{cen} and σ_{sur} and the corresponding coefficients of the fit are given in Table S1 and S2 for each neuron type.

Temporal filter model

The time-reversed impulse responses shown in Figure 2&4 are cross-sections through the center of the spatiotemporal receptive fields along the time axis and are averaged over the three center pixels. For the frequency domain representations in Figure 2&4, impulse responses were Fourier-transformed, averaged, and the resulting amplitude spectrum (absolute value) was divided by the power spectrum of the stimulus for frequencies below 5.5 Hz (below the Nyquist frequency). This is equivalent to deconvolving the impulse response with the stimulus auto-correlation and thereby correcting for non-white input signals [S7]. All frequency-space-representations are plotted on a double logarithmic scale expressing all signal gains in decibel according to convention in filter theory.

The complicated relationships between calcium, calcium indicator, voltage and neurotransmitter release of a cell render it impossible to precisely characterize each of these aspects having access to only the calcium indicator fluorescence as a read-out. However, we can assume under certain conditions that the calcium indicator itself essentially acts as a simple low-pass filter on the calcium signal [S10], which is a kind of distortion that we are able to correct for by applying deconvolution.

GCaMP6f is designed to have especially fast kinetics. However, we can find decay constants in the order of several hundreds milliseconds that vary depending on the system under observation [S1]. As an approximation we chose a time constant of 350 ms for a plausible low-pass filter that distorted the calcium signals in our system, which lies in the range of reported decay constants for GCaMP6f [S1, S11].

We corrected the frequency domain representations of the temporal filters of all cells by dividing the spectra with the frequency response of a 1st order low-pass filter with this time constant. Since this was restricted to frequencies below the Nyquist frequency, we did not have to apply additional techniques to avoid the impact of poor signal-to-noise ratios at higher frequencies.

For quantitative description and further simulations, we sought to describe the response characteristic of each cell under each condition with a simplified model that catches the main properties. For that, we fitted simple 1st order filters to the corrected frequency responses of all cells. We did this separately for each condition, i.e. for control and CDM condition and for the raw filters (corrected by the stimulus power spectrum only) and the deconvolved filters (corrected by the GCaMP filter) respectively.

In particular, we approximated Mi1, Tm3, Tm1, Tm2 and Tm4 as band-pass filters and fitted a band-pass model consisting of a 1st order high-pass and a 1st order low-pass filter to the frequency responses (Figures S5, S6). The band-pass model was parametrized by a multiplicative amplitude and the two time constants of the filters. Parameters were optimized using a standard implementation of the Levenberg-Marquardt algorithm (SciPy). Similarly, Mi4, Mi9 and Tm9 frequency responses were fit using a 1st order low-pass filter model.

Computational modeling

Neural simulations (Figure 5 and 6) were based on a motion detector that combines preferred-direction enhancement and null-direction suppression, resembling a hybrid of a Hassenstein-Reichardt half-detector and a Barlow-Levick detector, as suggested in [S12].

Stimuli were simulated in a 2-dimensional space covering 90° in both azimuth and elevation with 1° resolution. Each hypothetical motion (half-)detector had three neighboring input lines (termed A, B and C) which were offset by 5° from each other along the horizontal axis (for simplicity). Each input line consisted of a spatial and a temporal filter that was applied to the stimulus before further processing. The spatial filter was modeled as a 2D convolution with a Mexican hat filter kernel using the above definition (see “Spatial receptive field model”) and the fitted parameters from table S1 and S2. The temporal filter consisted of either a 1st order band-pass or as a 1st order low-pass filter with the time constants from the table correspondingly. Subsequent rectification simulated the polarity selectivity of the input lines to the downstream motion detector. To implement the nonlinear interaction between the three input lines in the most simplified, we modelled the nonlinear action as $A \cdot B / (C + 0.1)$ involving only one free parameter to avoid division by zero. 270 of these elementary motion detectors were arranged on a 2-dimensional grid, separated by 5° from each other.

To evaluate the performance and tuning of the simulated detectors across stimulus frequencies, we measured the mean response of the simulated (half-)detectors to moving gratings at different speeds. Vertically oriented square wave gratings of 24° wavelength were swept over the detector array with 50 different velocities corresponding to 50 different contrast frequencies

logarithmically spaced between 0.01 Hz and 20 Hz. The gratings moved for 5s to the right followed by a pause of 0.5s and 5s of motion in the opposite direction. The time step for all simulations was 10 ms. The direction of the stimulus that elicited the strongest response across all frequencies was termed the preferred direction (PD) of the respective motion detector. Consequently, the other direction was the null direction (ND).

The direction selectivity of the resulting tuning curve was evaluated by defining a direction selectivity index (DSI)

$$DSI = \frac{\sum PD - \sum ND}{\sum PD + \sum ND}$$

where the sum goes over all frequencies simulated. This definition produces DSI values between 0 and 1, where 1 means perfect, and 0 means no direction selectivity. Secondly, the optimal frequency f_{opt} was defined as the stimulus frequency that elicited the strongest response in PD direction.

The above measure only quantifies the response difference between the two opposing directions of motion along the main axis of the detector. However, it cannot distinguish between detectors that differ in their response properties to intermediate directions of motion. Hence, we additionally assessed the directional tuning specificity of each detector by measuring its response to differently oriented moving gratings. We stimulated the model with square wave gratings of 24° wavelength, rotated by different angles from 0° to 360° in steps of 30° , and measured the mean response of the detector array at the optimal frequency f_{opt} , as determined above. From the corresponding simulated responses, the direction selectivity was quantified as the length of the normalized response vector:

$$L_{dir} = \left| \frac{\sum_{\varphi} \vec{v}(\varphi)}{\sum_{\varphi} |\vec{v}(\varphi)|} \right|$$

where $\vec{v}(\varphi)$ is a vector proportionally scaled with the mean detector response and pointing in the corresponding stimulus direction of motion given by the rotation angle φ of the stimulus. This quantity L_{dir} has been suggested as a robust measure of direction selectivity that includes both relative response magnitude and tuning width of a direction selective neuron [S13].

For the bar plots in Figure 5B&E (right column) the simulations were repeated also for all possible implementations of a two-arm detector whose nonlinear interaction was either modelled as $A \cdot B$ for a classical Hassenstein-Reichardt-(half-)detector or as $B/(C + 0.1)$ for a Barlow-Levick-detector.

All simulations were performed using Python 2.7.

Statistics

Throughout this article, values are reported as mean $\pm$ standard error (SEM). In order to quantify the significance of the effect of CDM application on the temporal response characteristics of the medulla cells, we defined three different measures for the impulse responses: a) the time to the first peak $t_{peak}^{(1)}$ is the time between the onset of the impulse response (defined as the time when it has reached 15% of its maximum value) and the time when it has reached its maximum value; b) the time to the second peak $t_{peak}^{(2)}$ is similarly defined as the time between the onset of the impulse response and the peak of the subsequent undershoot or overshoot, which is defined only for the band-pass filters; c) lastly, we defined a peak width w_{peak} as the width of the first

peak at half maximum. We quantified these values for each fly and tested the change between control and CDM condition for significance using a paired t-test.

Supplemental References

- S1. Chen, T.-W., Wardill, T. J., Sun, Y., Pulver, S. R., Renninger, S. L., Baohan, A., Schreiter, E. R., Kerr, R. A., Orger, M. B., Jayaraman, V., et al. (2013). Ultrasensitive fluorescent proteins for imaging neuronal activity. *Nature* 499, 295–300.
- S2. Strother, J. A., Wu, S.-T., Wong, A. M., Nern, A., Rogers, E. M., Le, J. Q., Rubin, G. M., and Reiser, M. B. The emergence of directional selectivity in the visual motion pathway of *Drosophila*. *Neuron in press*.
- S3. Maisak, M. S., Haag, J., Ammer, G., Serbe, E., Meier, M., Leonhardt, A., Schilling, T., Bahl, A., Rubin, G. M., Nern, A., et al. (2013). A directional tuning map of *Drosophila* elementary motion detectors. *Nature* 500, 212–216.
- S4. Ammer, G., Leonhardt, A., Bahl, A., Dickson, B. J., and Borst, A. (2015). Functional specialization of neural input elements to the *Drosophila* ON motion detector. *Curr Biol* 25, 2247–2253.
- S5. Pologruto, T. A., Sabatini, B. L., and Svoboda, K. (2003). ScanImage: flexible software for operating laser scanning microscopes. *BioMed Eng OnLine* 2, 13.
- S6. Reiser, M. B., and Dickinson, M. H. (2008). A modular display system for insect behavioral neuroscience. *J Neurosci Methods* 167, 127–139.
- S7. French, A. S. (1976). Practical nonlinear system analysis by Wiener kernel estimation in the frequency domain. *Biol Cybern* 24, 111–119.
- S8. Dayan, P., and Abbott, L. F. (2005). *Theoretical Neuroscience*. MIT Press.
- S9. Clark, D. A., Bursztyn, L., Horowitz, M. A., Schnitzer, M. J., and Clandinin, T. R. (2011). Defining the computational structure of the motion detector in *Drosophila*. *Neuron* 70, 1165–1177.
- S10. Borst, A., and Abarbanel, H. D. I. (2007). Relating a calcium indicator signal to the unperturbed calcium concentration time-course. *Theor Biol Med Model* 4, 7.
- S11. Schnell, B., Weir, P. T., Roth, E., Fairhall, A. L., and Dickinson, M. H. (2014). Cellular mechanisms for integral feedback in visually guided behavior. *Proc Natl Acad Sci USA* 111, 5700–5705.
- S12. Haag, J., Arenz, A., Serbe, E., Gabbiani, F., and Borst, A. (2016). Complementary mechanisms create direction selectivity in the fly. *eLife* 5:e17421.
- S13. Mazurek, M., Kager, M., and Van Hooser, S. D. (2014). Robust quantification of orientation selectivity and direction selectivity. *Front Neural Circuits* 8, 92.